Viscereality: Bioresponsive Virtual Reality for Box Breathing and Altered-State Experience

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Abstract

Breath-based interactions in virtual reality typically map breathing onto discrete visual objects or physiological proxies. Viscereality instead uses a particle field to translate internal physiological state into an exteroceptive representation, mapping lung volume onto the surrounding space, so that inhalation expands the environment and exhalation contracts it. Unlike discrete objects, particle representations have no fixed shape or boundary, and carry information through the position, motion, colour, and size of individual units and through their interactions, such as synchronization with nearest neighbours. Here we investigate a VR system developed to augment breathing practices and modulate altered states of consciousness by making these dynamics more fluid and synchronous during breath retention, and whether users are aware of this. In a preregistered within-subjects study, participants completed three ten-minute audio-guided box-breathing sessions in counterbalanced order: a high-symmetry VR condition in which particles transiently aligned into geometric coherence during breath holds, an asymmetric condition in which they remained irregular, and a dark-screen control. Subjective experience was assessed with an altered states of consciousness questionnaire and pictographic self-related measures. Relative to the dark-screen control, VR conditions improved breathing synchronization, increased SFoRC ratings corresponding to a greater perceived extension of the self in space, and produced altered-state profiles with strong perceptual, moderate pleasant, and weak distressing effects. The symmetry manipulation registered implicitly in participants' experience without supporting explicit identification of which session was symmetric. After a single ten-minute session, perceptual alterations and pleasant experiences were comparable to those reported for hyperventilation-based breathwork of longer duration, while distress was lower.

Keywords: biofeedback, breath-based interaction, breathwork, altered states of consciousness, particle systems

1 Introduction

1.1 Breath as an Affective Sensory Modality

Breathing is a unique and multifaceted sensory modality, as it carries both interoceptive and exteroceptive streams of information. One stream concerns internal sensory information about respiratory mechanics and blood-gas status, including signals related to lung inflation, airway and respiratory-muscle activity, and blood O_2/CO_2 balance (Chan et al. 2024; Engelen et al. 2023; Weng et al. 2021). Because inhaled air stems from the external environment, breathing also samples exteroceptive information about ambient conditions, including odours, airflow, and air temperature, through its coupling with orthonasal olfaction and thermo- and mechanosensory receptors in the upper airways (Mori and Sakano 2022a,b; Zhou et al. 2011; Sabnis et al. 2008; Kytikova et al. 2021). Breathing is also closely tied to affective experience. People sigh with relief after stress (Balban et al. 2023), hold their breath during freeze responses (Boiten et al. 1994), and hyperventilate during fear or panic (Jerath and Beveridge 2020; Kreibig 2010). Arousing negative emotions such as fear and anger are linked to shallower, more rapid breathing, whereas relaxation and contentment correspond to slower, deeper respiratory patterns (Boiten et al. 1994; Homma and Masaoka 2008; Jerath and Beveridge 2020). Unlike most visceral signals such as heartbeat, breathing is also amenable to voluntary control (Sarkar 2017), making this relationship bidirectional. Voluntarily changing respiratory patterns can modulate affective state (Jerath and Beveridge 2020; Ashhad et al. 2022), a principle that underlies breathwork as a therapeutic modality (Fincham et al. 2023; Balban et al. 2023; Siebieszuk et al. 2025). Breathwork techniques range from slow-paced protocols that target relaxation, stress regulation, and improvements in heart rate variability through sustained cardiovascular-respiratory coupling (Balban et al. 2023; Ahmed et al. 2021; Siebieszuk et al. 2025), to high-ventilation protocols such as cyclic hyperventilation that can facilitate more pronounced alterations in affect and consciousness (Bahi et al. 2024a; Lewis-Healey et al. 2024; Havenith et al. 2025a), effects increasingly posited as therapeutic, for instance in facilitating emotional breakthroughs (Fincham et al. 2025, 2024). Breathwork is already established as a therapeutic modality for stress, anxiety, and depression (Fincham et al. 2023; Bentley et al. 2023; Banushi et al. 2023), and a growing number of virtual-reality applications also incorporate breathing for therapeutic and health-related purposes (Cortez-Vázquez et al. 2024; Pancini et al. 2025). Yet this literature has so far explored only a relatively limited portion of the broader design space that breathing affords as an interaction modality in VR.

1.2 Breath-Based Interactions in VR

Existing breath-based VR systems span meditation (Prpa et al. 2020; Shamekhi and Bickmore 2018), anxiety regulation (Bossenbroek et al. 2020), stress management through heart-rate-variability biofeedback (Blum et al. 2019; Rockstroh et al. 2019), and rehabilitation (Shi et al. 2023; Alhammad 2024). In the context of our framework, we distinguish between two motifs in the design space regarding the mapping of the user's respiratory signal onto discrete visual objects, in contrast to environmental mappings. Object-oriented approaches typically place respiratory feedback in a discrete visual target. Third-person variants let the user watch an external entity respond to breathing, such as a cloud that approaches during inhalation and recedes during exhalation (Tinga et al. 2019). In first-person variants, the mapping is applied to a virtual body seen from the user's own vantage point. The "embreathment" illusion demonstrated that synchronising a virtual avatar's chest movements with the user's breathing enhances body ownership and agency (Monti et al. 2020a), and mapping respiratory signals onto the expansion and contraction of a virtual avatar bubble viewed from a third-person perspective has been found to enhance agency and body awareness (Wald et al. 2023). What these object-oriented designs share is that respiratory feedback is concentrated in a single visual target, whether an external creature or a self-representation, which the user attends to as an indicator of their breathing. These designs are typically evaluated in terms of embodiment, measuring whether the temporal coupling between breathing and visual response enhances body ownership, felt agency, and awareness of the respiratory signal (Monti et al. 2020a; Wald et al. 2023).

We contrast these with designs that embed respiratory feedback in the surrounding virtual environment rather than in a discrete object. DEEP (Bossenbroek et al. 2020; van Rooij et al. 2016) and Flowborne (Rockstroh et al. 2021) use diaphragmatic breathing to drive locomotion through immersive landscapes, rewarding paced breathing with movement and environmental change (Bossenbroek et al. 2020; Rockstroh et al. 2021). In a seated variant, Blum et al. (2020) mapped exhalation onto colour changes in flowers, trees, and rocks, keeping the feedback environmental but locally anchored to scene objects. A further group of environmental designs couples not the breathing signal directly but a slower downstream variable, typically heart rate variability, to ambient scene properties. In Blum et al. (2019) and *Heart Garden* (Rook et al. 2026), increasing coherence or heart rate variability gradually transforms the scene. These designs create calming feedback loops, but because the environment changes through a slower physiological proxy rather than respiration itself, moment-to-moment respiratory agency is attenuated. The locomotion-based designs provide immediate agency, but mainly as a progression mechanic rather than as a way of making surrounding space itself feel breath-coupled. In our design, we sought to synthesize these two approaches by mapping the user's breathing state onto an environment that expands and contracts with respiration. This provides immediate feedback and a sense of agency while presenting breathing as an ambient property of the surrounding space rather than as a distinct object requiring continuous attention. Specifically, the environment is composed of particles that respond to the user's breath such that inhalation expands the space outward and exhalation contracts it inward. In this way, changes in internal lung volume are mapped onto changes in external space. This mapping is closely related to what Goral et al. (2024b) describe as exteroceptive–interoceptive sensory substitution, in which internal bodily signals are translated into external sensory events that remain experientially linked to the body rather than appearing as separate readouts. By mirroring the felt expansion and contraction of the thorax, our system was intended to preserve the natural structure of breathing experience and thereby function as an intuitive interface that can be understood with minimal explicit instruction (Antle et al. 2009). This choice was also informed by evidence from embodied cognition showing that people judge quantities as larger after inhalation than after exhalation (Belli et al. 2021), which may indicate a possible, though still speculative, link between respiratory state and the perception of space. More broadly, this embodied metaphor appears across a range of immersive biofeedback systems that map respiration onto expanding and contracting visual stimuli (Wald et al. 2023; Goral et al. 2024b; Barton et al. 2024; Stepanova et al. 2020). Two existing systems are conceptually closest to the approach presented here. Barton et al. (2024) placed participants inside a virtual orb covered in abstract fractal patterns that expanded and contracted as participants spread and closed their arms in synchrony with their breathing; however, the interaction was structured as an explicit movement-synchronisation task rather than as an environment that responds to breathing on its own. Goral et al. (2024b) introduced exteroceptive–interoceptive sensory substitution in a system that translated respiratory signals into dynamic visual and auditory stimuli intended to feel continuous with the breathing body; however, their implementation externalized the signal through projection rather than embedding it in an immersive three-dimensional environment. Our system was designed to combine the strengths of these two approaches by pairing real-time environmental responsiveness that affords immediate respiratory agency with feedback distributed across surrounding space rather than concentrated in a single object or task. Rather than relying on semantic cues such as naturalistic scenery, skyboxes, or rendered objects, we pursued a bottom-up design in which the sense of surrounding space emerges from relations among many simple particles, drawing on Gestalt principles such as proximity, common fate, and continuity (Wagemans et al. 2012) so that their dynamic arrangement, and specifically their responses to expansion and contraction, conveys depth and closeness dynamically within an abstract particle-based environment. This design rationale also motivates the broader question of whether coupling an internal bodily signal such as respiration to the surrounding sense of space can also reshape the sense of self, which we consider next through a review of relevant VR experiments.

1.3 VR interventions testing the malleability of the self

Virtual reality has been used extensively to investigate the malleability of bodily self-experience through multisensory manipulation. In full-body illusion paradigms, synchronous visuotactile stimulation between a participant's physical body and a seen virtual body can shift perceived self-location and induce a sense of ownership over the virtual body (Lenggenhager et al. 2007; Petkova and Ehrsson 2008). Later work showed that interoceptive signals can serve a similar synchronising function. Presenting a virtual body or silhouette that flashes in synchrony with the participant's heartbeat enhances body ownership and alters self-location even in the absence of tactile stimulation (Aspell et al. 2013; Suzuki et al. 2013). Respiratory signals have been used in a similar way. For example, synchronising a virtual avatar's chest movements, viewed from a first-person perspective, with the user's breathing enhances body ownership (Monti et al. 2020a). Similarly, mapping respiratory signals onto the expansion and contraction of a virtual bubble, viewed from a third-person perspective, enhances agency and body awareness (Wald et al. 2023). Collectively, these studies show that a range of sensory modalities, including touch, breathing, and cardiac activity, can be used to reinforce visual feedback and thereby strengthen the sense of embodiment toward a virtual object or avatar from both first-person and third-person perspectives (Lenggenhager et al. 2007; Petkova and Ehrsson 2008; Aspell et al. 2013; Suzuki et al. 2013; Monti et al. 2020a; Wald et al. 2023).

A related but distinct line of research concerns peripersonal space, the region immediately surrounding the body where the brain integrates multisensory signals in body-centered coordinates to maintain a dynamic boundary between self and environment (Serino 2019; Bufacchi and Iannetti 2018). This boundary is not fixed; it extends with tool use (Maravita and Iriki 2004), shifts with emotional valence and social context (Bufacchi and Iannetti 2018; Serino 2019), and interacts with interoceptive aspects of bodily self-representation (Tsakiris et al. 2011a; Azzalini et al. 2019). Recent virtual-reality work further suggests that embodiment and multisensory manipulations can reshape peripersonal space in VR (Petrizzo et al. 2024; Karakoc et al. 2025). While the embodiment paradigms reviewed above manipulate the felt relationship between self and a visual body representation, peripersonal space research highlights that self-experience also depends on how the brain represents the spatial relationship between the body and the surrounding environment (Serino 2019; Bufacchi and Iannetti 2018).

Notably absent from both lines of research are paradigms that intervene through the surrounding environment rather than through a representation of the body itself. In most VR studies of embodiment and self-location, self-related experience is altered by manipulating how the body is represented, with any effects on perceived space arising indirectly through changes in how the body is seen in space (Guy et al. 2023). Although VR research has also developed techniques that manipulate space itself, such as scaling, translation, and redirected movement, these are typically used to support navigation and interaction rather than to investigate self-experience (Abtahi et al. 2022). By contrast, the present approach couples respiration not to a virtual body or discrete object, but to the spatial extent of the surrounding environment itself. This may engage the boundary between body-centred and environment-centred space in a way that body-focused paradigms do not.

There is a broad consensus that the sense of self is multidimensional (Millière 2020; Taves 2020), and the present study assessed three complementary pictographic measures. These were body-boundary salience (Dambrun 2016), spatial frame of reference (Hanley and Garland 2019), and perceived self-size (Vidal et al. 2025). At the same time, the intervention was expected to affect not only self-experience, but potentially the broader structure of subjective experience. The present study therefore also considered its effects in relation to psychometric frameworks from altered-states research, which informed both measurement and the design rationale of the VR system described next.

1.4 Breathwork as an experiential design space for virtual reality

Embedding a breathing protocol in a responsive virtual environment introduces specific design constraints. Different breathing practices produce different respiratory patterns, including variations in the pace, duration, and force of inhalation and exhalation. Visual feedback coupled to these dynamics must therefore avoid becoming dizzying, claustrophobic, or overwhelming. Visual feedback coupled to either level must avoid becoming disorienting. In VR, large-amplitude radial scene motion, particularly in the visual periphery, is a documented source of cybersickness (Chang et al. 2020; Pöhlmann et al. 2021), and contracting environments can elicit involuntary defensive postural responses even in non-phobic users (Kodithuwakku Arachchige et al. 2022). These risks are compounded when the user is in a heightened or unfamiliar experiential state, given that negative affect and arousal are associated with increased VR sickness severity (Kaufeld et al. 2022). The choice of breathing protocol therefore shapes not only what physiological and experiential effects to expect, but also what the visual environment can safely and comfortably do in response.

High-ventilation breathwork practices such as holotropic breathing, conscious connected breathing, and cyclic hyperventilation involve sustained increases in breathing rate and depth that, over extended sessions, induce pronounced physiological changes, including respiratory alkalosis and reductions in cerebral blood flow, and can give rise to marked changes in subjective experience (Lewis-Healey et al. 2024; Kartar et al. 2025; Fincham et al. 2024). Participants report experiences of oceanic boundlessness, in which the felt boundary between self and world dissolves into a sense of unity or merging with the surroundings; visionary restructuralisation, involving vivid geometric imagery, complex visual scenes, and synesthetic blending of sensory modalities; and dread of ego dissolution, characterised by a felt loss of control over one's thoughts and sense of identity (Bahi et al. 2024a; Havenith et al. 2025a; Lewis-Healey et al. 2024). These experiential dimensions overlap with those reported under medium- to high-dose psilocybin as measured by the 11D-ASC (Studerus et al. 2010; Havenith et al. 2025a), and are often accompanied by emotional catharsis and strong bodily sensations such as tingling, dizziness, and autonomic arousal (Havenith et al. 2025a; Lewis-Healey et al. 2024; Fincham et al. 2024, 2025). However, the rapid, large-amplitude chest excursions characteristic of high-ventilation protocols would produce correspondingly rapid expansion and contraction of a breath-coupled particle environment, and the accompanying physiological arousal and altered experiential state would coincide with precisely the visual conditions most likely to provoke cybersickness and spatial discomfort. For these reasons, we chose to facilitate slow-paced breathing rather than high-ventilation breathwork in the present system.

Slow-paced breathing occupies a different region of this subjective and affective space. Its established effect profile is characterised by relaxation, reduced anxiety, increased comfort, and calm alertness, supported by parasympathetic dominance and enhanced heart rate variability (Zaccaro et al. 2018; Balban et al. 2023; Fincham et al. 2023). Slow breathing is also one of the most common attentional anchors in guided meditation practice (Gerritsen and Band 2018; Zaccaro et al. 2018), and evidence from experienced pranayama practitioners indicates that very slow nasal breathing can induce altered states of consciousness characterised by altered body perception, distortions in time sense, unusual salience, increased joy, and reduced anxiety (Zaccaro et al. 2022). The mild but documented non-ordinary experiential profile of slow-paced breathing provided a low baseline against which any augmentation by the visual environment would be easier to detect, while keeping the environment's spatial dynamics within a comfortable perceptual range.

Augmenting subjective experience is clinically relevant because experiential intensity during altered-state interventions predicts therapeutic outcomes (Roseman et al. 2018; Romeo et al. 2025; Havenith et al. 2025a). We chose a slow-paced protocol precisely because its mild baseline provides sensitivity for detecting whether we can enhance specific attributes of the subjective experience through VR. To characterise this experiential profile and compare it with existing breathwork and psychedelic literatures, we adopted the 11D-ASC (Studerus et al. 2010; Dittrich 1998), the standard instrument across these fields. Its subscales capture the phenomenological features (e.g., boundary dissolution, changed imagery, synesthetic blending) that a breath-coupled environment might selectively amplify, and can be summarised into three higher-order dimensions reflecting positive, distressing, and perceptual effects (Stocker et al. 2025). Because we also examine how these dimensions evolve over the course of a breathing session, we include psychometric time-series ratings, which we refer to as temporal experience tracers (Jachs 2021). We match the tracers one-to-one with the 11D-ASC dimensions to compare global ratings with time-resolved intensity profiles of the same experiential constructs. The motivation to enhance subjective experience is also reflected in the aesthetic design of the visual environment, which we describe next.

1.5 Particle dynamics as a bioresponsive design medium

Viscereality is a bioresponsive VR system developed through an iterative design process, parts of which have been reported previously with respect to its design rationale, technical implementation, and interaction mechanics in human-computer interaction contexts (Fejer et al. 2025; Fejer 2026). The present study provides its first controlled empirical evaluation, testing whether the system's particle dynamics measurably shape subjective experience during breathwork. As described above, the system maps lung volume onto a surrounding particle field through interoceptive-exteroceptive sensory substitution (Goral et al. 2024b). Inhalation expands the environment outward, whereas exhalation contracts it inward. Existing implementations of this type of coupling typically use modelled surfaces, either presented as localized targets linked to the user (Wald et al. 2023) or projected onto the walls of a physical room (Goral et al. 2024b). In both cases, the user watches a representation change size, but a uniform surface that grows or shrinks does not, on its own, strongly convey that the surrounding space itself is approaching or receding; that impression depends on motion-based depth cues that arise when many elements move relative to one another (Rogers and Graham 1979). We designed Viscereality as a particle system specifically to afford this. The environment is composed of approximately 2,500 particles distributed on a geodesic icosphere, and the sense of spatial expansion and contraction arises from Gestalt grouping among these elements, specifically proximity, common fate, and continuity (Wagemans et al. 2012), rather than from a texture mapped onto a surface. Because each particle independently carries multiple visual properties (position, size, colour, transparency, motion trajectory), the same elements that produce the breath-coupled spatial dynamics can simultaneously encode additional biofeedback channels, such as heart rate variability coherence (Fejer 2026), without requiring separate visual layers. And because inter-particle interaction rules, specifically oscillator coupling across neighbours, generate emergent aesthetic properties such as dynamic symmetry and geometric coherence, these can be parameterised independently of the breathing mechanics, making the aesthetic texture of the environment an experimentally controllable variable. Throughout the breathing cycle, the particle field remains continuously active. Surface-wave oscillations, chromatic cycles, and shape fluctuations continue during inhalations, exhalations, and both breath-hold phases. The within-VR manipulation therefore does not switch visual motion on or off; rather, during breath holds the high-symmetry condition transiently aligns these ongoing particle cycles into coherent geometric organisations, whereas the asymmetric condition leaves the same dynamics desynchronised. To our knowledge, no existing interoceptive-exteroceptive sensory substitution system takes this particle-based approach. Aesthetics in VR are often treated as incidental to intervention design, yet recent work identifies them as among the strongest predictors of emotional response and experience quality in immersive environments (Marcolin et al. 2021; Cheiran et al. 2025). In Viscereality, aesthetics are not a cosmetic layer but emerge from the same particle dynamics that carry the respiratory coupling, drawing on a longer history of expressive and aesthetic design in particle-based systems applied in visual effects animation.

1.5.1 Particle systems as a generative medium

Particle systems were introduced by Reeves (1983) to represent visual phenomena that have no fixed boundary and no stable shape, including fire, smoke, clouds, and water. In his formulation, each particle is born with stochastically assigned properties (position, velocity, colour, size, transparency, lifetime), moves and changes over time, and eventually vanishes. The overall form is never fully specified; it emerges from the aggregate of many simple elements, each governed by a small set of controllable parameters. This is the property that makes particle systems useful as a bioresponsive medium because each particle independently carries multiple visual channels, so a single population of elements can simultaneously respond to physiological signals through spatial position, colour shifts, size, transparency, or motion trajectory without requiring separate visual layers for each mapping. Building on this work, Reynolds (1987) introduced interaction between elements, enabling coordinated collective behaviour to emerge from local neighbour rules, whereas in a standard particle system each element responds only to global parameters and physics without awareness of its neighbours. Reynolds specified three local rules. They were to avoid collision with nearby elements, match their velocity, and maintain proximity to them. This became foundational for the ability of 3D animation to produce the fluid, coordinated group motion recognisable in flocking birds, schooling fish, or stampeding herds, all arising bottom-up from neighbour interactions without a centrally scripted choreography.

1.5.2 Particle systems as an affective medium

Research on abstract emotion representation shows that affective meaning can be conveyed through low-level perceptual parameters of abstract expressions (De Rooij et al. 2013). In particular, affective valence can be mapped through semantically minimal features such as motion speed and trajectory smoothness (Bartram and Nakatani 2009; Pollick et al. 2001; Burger and Toiviainen 2020), angular or rounded shape contours (Blazhenkova and Kumar 2018; Aronoff 2006), and line orientation (Damiano et al. 2023). In conceptual terms, Khatin-Zadeh et al. (2019) argue that motion provides an effective vehicle for representing emotion because both unfold dynamically over time, allowing changes in affective state to be expressed through movement, direction, and position in space. This claim has also been borne out empirically in swarm robotics, where the collective motion of distributed abstract disc-shaped agents has been shown to alter the observer's state of flow and sense of time relative to observing a single moving agent (Kaduk et al. 2024). Dietz et al. (2017) found that higher speed in swarm motion increased perceived emotional intensity, while smooth trajectories, relative to jittery motion, were associated with more positive affect. Extending this approach, Santos and Egerstedt (2021) translated trajectory and motion into swarm choreographies for Ekman's six basic emotions (Ekman 1993). Rounded and smooth configurations marked positive or low-arousal states, while angular, clustered, or dispersive formations with fast or slow trajectories marked negative and higher-arousal states. Participants identified the intended emotions above chance (Santos and Egerstedt 2021). What makes these findings from swarm robotics relevant here is that their expressive effects arise from relational motion among many simple elements rather than from the form of any individual agent. The same principle holds in *Viscereality*, where local coupling among particles governs trajectory speed, smoothness, synchrony, clustering, and dispersal, producing higher-order collective motion patterns from which affective qualities may emerge. The particle field can therefore be treated as a medium for conveying affective states through abstract visual dynamics rather than semantic or figurative cues.

1.5.3 Particle systems as an aesthetic medium

Beyond serving as a medium for affective mapping, we designed the particle field as an aesthetic means of rendering affective states through abstract spatiotemporal organization. Research on abstract visual art suggests that non-figurative forms evoke aesthetic and affective responses through the isolation of simple features, the grouping of many elements into emergent wholes, and the amplification of perceptual qualities through Gestalt grouping principles (de Rooij et al. 2013; Ramachandran and Hirstein 1999). In our particle system, these same grouping principles also help convey depth when approximately 2,500 ring-shaped particles expand and contract around the user. Because each particle is visually minimal, higher-order forms are not pre-given but emerge through local relations among neighbouring elements, allowing circles, lattices, polygons, and other geometric organisations to arise through changes in three-dimensional location. The appearance of such forms can itself become aesthetically salient, as the resolution of an initially indeterminate field into a coherent Gestalt is known to produce pleasurable moments of insight or 'aesthetic Aha' moments (Muth and Carbon 2013; Spehar and van Tonder 2017; Spee et al. 2024). In Viscereality, this made it possible to translate oscillator-level control into changes in overall visual order, symmetry, and coherence without moving to figurative imagery.

Another design influence was the geometric and kaleidoscopic qualities of hallucinatory and psychedelic imagery, which are often marked by geometric patterns, bilateral symmetry, and perceptual ambiguity (Klüver 1966; Billock and Tsou 2012; Lawrence et al. 2022; Rubin et al. 2010). This motivated our use of ring-shaped particles, as their overlaps can give rise to distinctive emergent patterns during the breathing cycle. Because these qualities emerge from shifting relations among many simple particles rather than from fixed symbolic forms, they remain open in meaning and foreground a defining affordance of the medium, namely its capacity to continually reorganise and transform its aesthetic structure over time.

In the context of VR design, Glowacki (2024) terms such non-figurative scenes a weak representational aesthetic, in contrast to photorealistic depictions that render scenes and bodies as recognizable real-world entities. In practice, this aesthetic has been applied in VR scenarios designed to induce altered states of consciousness through the representation of participants' bodies as luminous, semi-corporeal avatar bodies composed of particle clouds with mergeable body boundaries, contributing to the malleability of self-experience and altered subjective states (Glowacki et al. 2020, 2022; Glowacki 2024; Vidal et al. 2025). Our approach adopts a related design aesthetic while focusing more specifically on diminishing the felt boundary between self and environment.

To evaluate the particle system's capacity to manipulate aesthetic properties, we operationalized symmetry as an additional independent variable that varied across conditions. Previous studies have linked visual symmetry to more positive affect and perceptual fluency (Bertamini et al. 2013; Pecchinenda et al. 2014; Makin et al. 2012), while synchrony in group movement has been linked to greater enjoyment and more positive affective appraisal (Vicary et al. 2017; Mogan et al. 2017). We therefore asked whether higher degrees of symmetry would enhance positively valenced experiences in the context of Viscereality (Kuhn 2024; Johnson 2023). We were also interested in whether participants would be able to detect and discriminate between the two conditions.

1.6 Present Study

The present study provides the first controlled empirical evaluation of Visceality with naive participants. We used a preregistered within-subjects design with $N = 39$. Each participant completed three 10-minute audio-guided box-breathing sessions in counterbalanced order. These were a high-symmetry condition, in which positive oscillator coupling transiently aligned ongoing particle oscillations into dynamic geometric coherence during breath-hold phases; an anti-symmetry condition, in which negative coupling produced a visually irregular particle field throughout; and a dark-screen control (see Section 2.3.4). The two particle conditions used identical particles, breath-coupled spatial dynamics, colour palette, and audio guidance, differing only in oscillator coupling parameters. The central within-VR contrast therefore concerned the aesthetic organisation of an otherwise identical breath-responsive scene.

We assessed four substantive outcome domains together with a methodological blinding assessment. Breathing adherence was quantified as the phase-locking value between the observed respiratory signal and an ideal reference trajectory. Self-related experience was assessed with three pictographic measures, namely body-boundary salience (Dambrun 2016), spatial frame of reference (Hanley and Garland 2019), and perceived self-size (Vidal et al. 2025). Altered states of consciousness were assessed retrospectively with the 11D-ASC (Studerus et al. 2010; Dittrich 1998). Blinding efficacy between the two active VR conditions was assessed retrospectively with a two-stage procedure. Before unmasking, participants completed 12 probe questions covering condition-specific, shared, and sham features. After being told that the two active scenes differed, they indicated in which order the high-symmetry and anti-symmetry scenes had been presented, together with confidence ratings. These responses were used to estimate condition-identification blinding with a standard single-group Bang blinding index and an exploratory confidence-weighted directional variant (Bang et al. 2004).

We hypothesised that, relative to the dark-screen control, the bioresponsive VR conditions would increase breathing adherence and altered-state intensity and would also influence the three preregistered self-related measures. Within VR, we further expected the high-symmetry condition to enhance positively valenced experiences and produce stronger alterations in self-related experience than the anti-symmetry condition. The preregistration did not specify a separate directional prediction for each pictographic outcome. We therefore treated the three outcomes as one family and controlled the false-discovery rate across them.

2 Methods

2.1 Participants

Participants were recruited from the University of Konstanz psychology student participant pool and received course credit for participation. The study used a within-subject design in which each participant completed all three conditions in one of six counterbalanced block-order permutations, with a preregistered target of 6 completers per permutation and 36 total. We did not systematically assess prior experience with virtual reality, meditation, breathwork, or psychedelics, so the sample should be understood as a general psychology participant-pool sample rather than as a formally screened novice cohort. We tested 47 participants; 8 were excluded based on breathing signal quality (phase-locking value (PLV) outlier criterion described below), yielding a final sample of $N = 39$ (6–7 per permutation), preserving approximately equal representation across all six condition sequences. Among the included participants, mean age was 20.8 years; 37 were female (mean age 20.7 years) and 2 were male (mean age 23.0 years).

2.2 Design, Conditions, and Procedure

Participants were seated in a bean bag, wore noise-blocking headphones, held a controller against their abdomen, and completed all breathing blocks, questionnaires, and outcome measures fully immersed in virtual reality. The three-session procedure was automated end-to-end in the headset: once the headset was put on, participants followed in-VR instructions for the breathing blocks and all post-block outcome measures while remaining seated, with minimal experimenter intervention beyond initial setup and debriefing. Before the experimental sequence, participants completed a 1-minute guided breathing practice block for acclimation, familiarizing them with controller placement and the paced-breathing rhythm. Each participant then completed three conditions, with session order assigned automatically by random selection from the six counterbalanced block-order permutations. Because the dynamic symmetry manipulation was embedded in the software-controlled scene logic and the condition sequence was assigned automatically at runtime, neither the participant nor the experimenter knew which condition would appear in which session. In all conditions, participants followed a 10-minute audio-guided box-breathing exercise (sourced from a publicly available recording) consisting of repeated 4-4-4-4 cycles with four seconds each of inhalation, breath hold, exhalation, and breath hold. In the high-symmetry condition, the bioresponsive particle environment formed dynamic symmetric patterns during breath-hold phases. In the asymmetric condition, visual features of the particles remained asymmetric to one another throughout. In the dark-screen control condition, the particle environment was not displayed; participants viewed a dark field containing only a central fixation marker that also provided breathing-performance feedback (Section 2.3.4). We refer to this condition as the “dark-screen control” throughout. After each block, participants completed the full assessment battery (questionnaires and pictographic scales) while remaining in virtual reality. Figure 1 summarizes the full procedure, including the repeated condition loop and blinding sequence, using the same condition-color mapping as the Results figures.

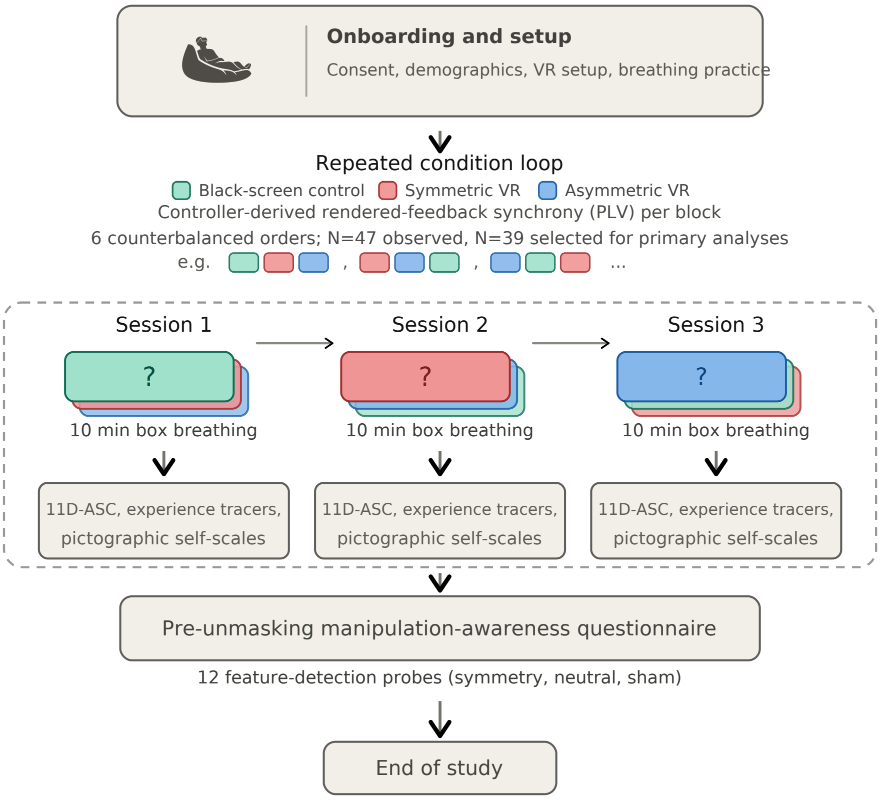

Fig. 1 Study procedure. After onboarding, each participant completed three 10-minute box-breathing sessions (dark-screen control, symmetric VR, asymmetric VR) in one of six counterbalanced orders, with the full assessment battery administered in VR after each block. A blinding questionnaire followed the third block before the study concluded.

2.3 Virtual-Reality Intervention

Vicereality is a bioresponsive virtual-reality system that maps respiratory signals to dynamic visual feedback within an immersive 3D particle environment (Fejer et al. 2025; Fejer 2026). A video summary of the system is available online.[1] The system runs on a Meta Quest 3 headset (standalone, untethered) and was built in Unity, with CPU-optimized parallel processing (Unity Job System with Burst compilation) maintaining a stable 72 Hz frame rate. The two active virtual-reality conditions used the same visual environment, audio guidance, and logging pipeline; they differed only in a single engine parameter controlling oscillator coupling dynamics. The dark-screen control used identical audio guidance and data logging but omitted the particle visualization. We focus here on the experiment-relevant functional logic; fuller technical

[1] https://www.youtube.com/watch?v=Qm-lmC-xYCI; archived at https://osf.io/4enfg

and implementation details of the Viscereality engine are described in the prior system papers (Fejer et al. 2025; Fejer 2026).

2.3.1 Breath Detection

Breath detection follows the approach introduced by Flowborne (Rockstroh et al. 2021), using the VR controller's positional tracking to infer respiratory displacement without additional sensors. During an initial calibration, the user places the controller against the lower abdomen to establish a tracking axis. Each frame, the algorithm reads the controller's position and orientation, projects motion onto this breathing-relevant axis, and updates a short-term versus long-term smoothed motion estimate. The difference between these smoothed signals functions as a breathing trend signal. Positive values indicate inhalation, negative values indicate exhalation, and near-zero values indicate a pause. Samples are rejected as tracking artifacts if controller rotation is too abrupt or if the inferred trend is implausibly large. The resulting classification (inhale, exhale, pause, or bad tracking) drives the visual feedback pipeline described below (Figure 2). Detailed implementation choices and parameterization are documented in the Viscereality technical descriptions (Fejer et al. 2025; Fejer 2026).

2.3.2 Sphere Radius and Visual Feedback

The user is positioned at the center of a large particle-covered sphere. The primary feedback loop couples sphere radius to breathing. Inhalation causes the sphere to expand outward, exhalation causes it to contract inward, and breath holds maintain the current radius. Functionally, the controller-derived breathing-state label is converted into a signed expansion/contraction command whose magnitude is scaled to the configured radius range and the intended inhale/exhale timing. That command updates a target sphere radius, and the rendered radius is then smoothed with a damped interpolation step (0.25 s smooth time; overshoot prevention enabled). In practice, this means participants do not see a raw controller-motion trace; they see a bounded, temporally stable visual signal that preserves breathing direction while suppressing jitter. Full implementation details of the radius controller are described in the prior Viscereality system reports (Fejer et al. 2025; Fejer 2026).

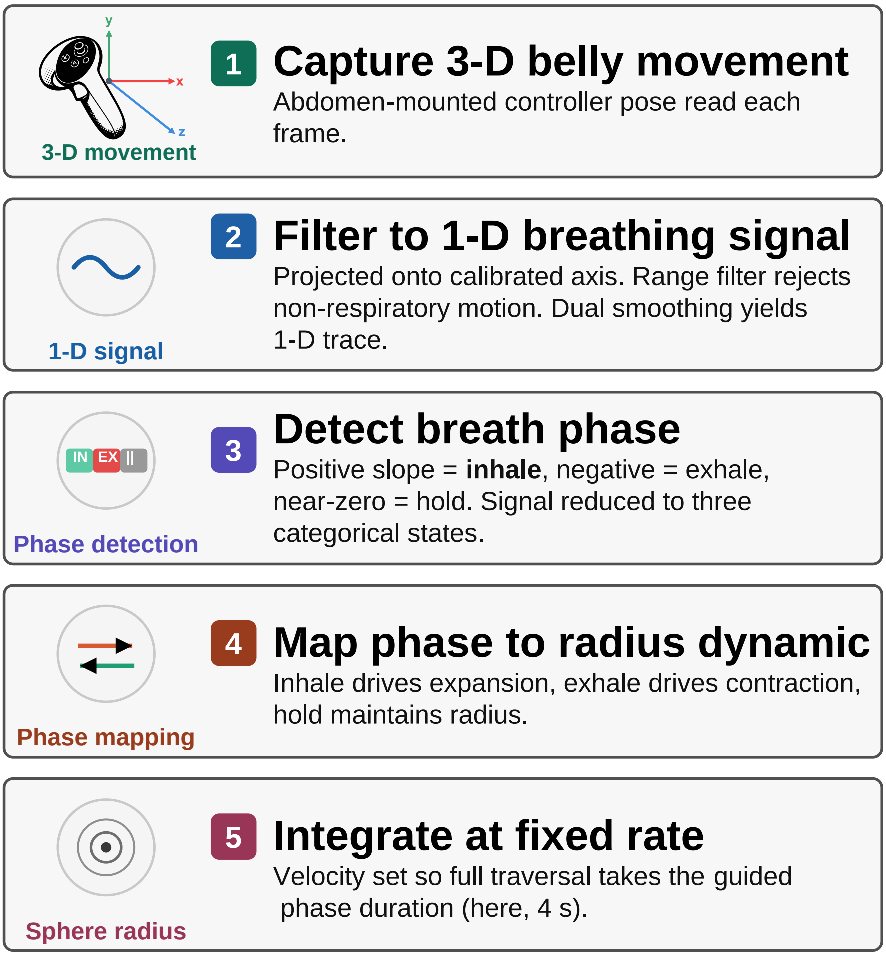

Fig. 2 Signal processing pipeline from abdomen-mounted controller motion to rendered sphere radius. Raw 3D movement is projected onto a calibrated breathing axis, classified into inhale, exhale, or hold phases, and mapped onto expansion and contraction dynamics that update the visible sphere at a fixed velocity.

The sphere surface is composed of approximately 2,500 view-aligned ring particles distributed on an icosphere mesh for uniform geodesic spacing. Each particle is rendered with a radial gradient that produces depth cues through overlap, and during expansion and contraction, particles generate cone-like tracer effects that reinforce the perception of three-dimensional movement. Surface wave oscillations, structured noise, and unsynchronized chromatic cycles maintain continuous visual dynamics across all breathing phases, preventing the environment from appearing static during breath holds.

2.3.3 Oscillator Network and Condition Manipulation

The second layer of the system implements a Kuramoto oscillator network across the sphere's vertices (Acebrón et al. 2005; Rodrigues et al. 2016). Each particle has an internal oscillation phase (analogous to a clock angle) and interacts with its neighbors through coupling weights that determine whether nearby particles tend toward synchronization or away from it. If the coupling weight is positive, neighbors are pulled toward shared timing (phase attraction). If negative, the interaction becomes *anti-coupling*, and neighbors are pushed away from alignment (phase repulsion).

The network uses a two-tier neighbor structure reflecting the icosphere topology. Most vertices have 6 immediate (tier-1) neighbors and 12 next-nearest (tier-2) neighbors, while 12 vertices at the original icosahedron poles have 5 tier-1 neighbors. At each update step, each oscillator advances according to its own intrinsic rhythm plus a neighborhood interaction term. In plain terms, the algorithm compares an oscillator's current phase with those of its neighbors and then adjusts it toward or away from them depending on the sign and magnitude of the tier-specific coupling weights. This is the standard Kuramoto logic (phase-offset-dependent attraction/repulsion), implemented here with two neighbor tiers and a global coupling gain, and updated in parallel each frame together with particle position, size, and color. The formal implementation and parameterization are described in the prior Viscereality publications (Fejer et al. 2025; Fejer 2026).

The two active conditions differed in how coupling weights were assigned. In the high-symmetry condition, tier-1 and tier-2 coupling weights were interpolated dynamically as a function of a coherence scalar (0–1), together with a coherence-dependent frequency scaling term. Verbally, low coherence pushes the system toward anti-coupled, irregular dynamics, whereas higher coherence progressively shifts interactions toward phase attraction and more synchronized motion. As a result, particle motion, color, and size periodically coordinate into coherent geometric patterns, such as flower-of-life-like lattices, during breath-hold phases before dissolving back into disorder when breathing resumes.

In the asymmetric condition, the system did not use this dynamic interpolation. Instead, coupling weights were held at fixed negative values (anti-coupling), so local interactions were permanently biased toward phase repulsion and the particle field remained visually irregular throughout. The manipulation was thus dynamical rather than geometric. The same scene, particles, and audio were used in both conditions, and only the oscillator interaction rules changed.

2.3.4 Breathing Performance Indicator

All three conditions displayed a minimal breathing performance indicator at the center of the participant's field of view. This consisted of a white pentagon ring whose individual line segments scaled with sphere radius: at half-volume (midpoint between full inhale and full exhale), the pentagon was fully visible; as the participant inhaled or exhaled toward the extremes, the segments progressively shortened and disappeared entirely at maximum and minimum volume. The indicator thus functioned as an unobtrusive loading-bar-style monitor of breathing excursion. Participants were instructed that the goal was to make the pentagon disappear completely on each inhalation and exhalation, confirming sufficient breathing depth.

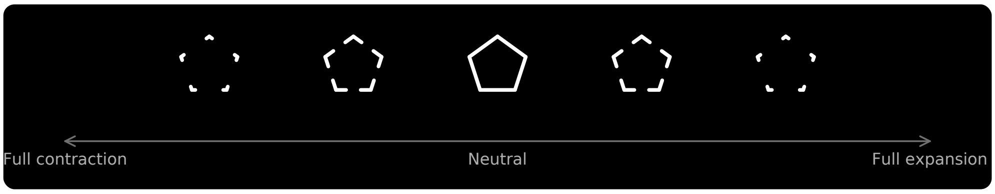

Fig. 3 Central breathing-performance indicator shown in all three conditions. The pentagon ring was fully visible at the neutral sphere radius and progressively disappeared toward full contraction or full expansion, providing condition-matched feedback about breathing excursion.

2.4 Outcome Measures

We retained the five preregistered measurement blocks used in the assessment battery, but organize them here into four recurring outcome families that also structure the Results section. These were adherence to box breathing, self-related measures, altered states of consciousness, and the blinding questionnaire. Within the altered-states family, we present the derived 3D-ASCr composites first as a dimensionality-reduced overview of the same 11D-ASC data, followed by the underlying 11D-ASC questionnaire.

2.4.1 Adherence to Box Breathing

Adherence quantifies how closely each participant's breathing followed the instructed 4-4-4-4 box-breathing rhythm across each 10-minute block. The preregistered plan was to compute this from discrete phase labels produced by the breath-detection algorithm (Section 2.3.1), which classifies each frame as inhale, exhale, hold, or bad tracking. The intended metric was phase-duration error, defined as the absolute deviation from the 4-second target per breathing phase, averaged across valid cycles per condition, with participant exclusion if fewer than 70% of cycles were valid or if mean error exceeded 2 standard deviations above the sample mean.

The recorded phase sequences did not yield the expected clean succession of 4-second phases. Rather than discrete blocks of inhale, hold, exhale, and hold, the recorded data consisted of highly fragmented segments interspersed with brief transitional or ambiguous intervals. Computing phase-duration error from this data required first reconstructing plausible box-breathing cycles from the raw phase stream, a step that was not anticipated at preregistration. We explored several reconstruction strategies, including rule-based merging of adjacent micro-segments and temporal smoothing algorithms, but each approach introduced its own arbitrary parameter choices, such as minimum segment duration thresholds, smoothing window widths, and rules for combining adjacent same-phase fragments, that could materially influence the resulting error estimates. Because these post-hoc binning decisions carried more researcher degrees of freedom than the original preregistered metric implied, we adopted an alternative approach.

We quantified adherence as the phase-locking value (PLV) between two continuous time series. One was the sphere-radius signal that participants actually saw as visual feedback during the breathing session. The other was an ideal reference radius trajectory derived from the audio-instruction timestamps for perfect 4-4-4-4 breathing. PLV measures how consistently the phase relationship between two signals remains stable over time. A value near 1 indicates that the participant's breathing closely tracked the instructed rhythm throughout the block. A value near 0 indicates little consistent alignment. This approach circumvents the phase-reconstruction problem entirely, because it evaluates alignment at the level of the continuous visual feedback signal rather than attempting to recover discrete phase boundaries from a fragmented controller trace. PLV thus captures the same underlying construct as the preregistered metric, namely how closely breathing tracked the instructed rhythm, while avoiding the measurement artifacts that any post-hoc phase-segmentation strategy would have introduced. Figure 4 illustrates the relationship between observed and reference trajectories for a representative participant; the fragmented phase-label structure visible in the lower bars of each panel motivates the PLV-based approach described above.

Because PLV does not decompose into per-cycle validity counts, the preregistered 70%-valid-cycle plus 2-SD mean-error exclusion rule could not be transferred directly. Outlier exclusion instead used a lower-tail global IQR rule on condition-level PLV values, elevated to participant-level exclusion if any condition was flagged. Supplementary Section S2 describes PLV computation and quality screening in detail.

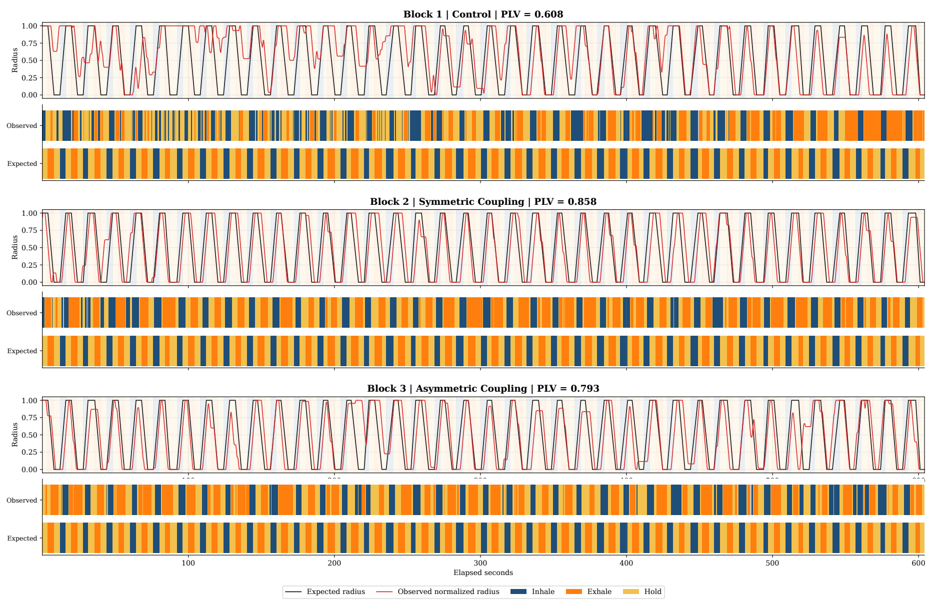

Fig. 4 Breathing dynamics for a representative participant (P01) across all three conditions. Observed sphere-radius traces (orange) are overlaid on the ideal 4-4-4-4 reference (blue), with per-block PLV values shown at right. Fragmented phase-label bars below each time series illustrate the segmentation noise that motivated the switch from phase-duration error to PLV as the adherence metric. Equivalent figures for all participants are provided in the OSF repository.

2.4.2 Self-Related Measures

Three single-item pictographic measures assessed self-related experience after each condition. The adapted Perceived Body Boundaries item (Dambrun 2016) presents seven silhouettes progressing from diffuse (A) to sharply delineated (G). Responses were mapped from A–G to 1–7, so higher analyzed scores indicate stronger boundary salience. For construct-facing exploratory correlations, the item was reverse-scored so that higher values indicate greater boundary dissolution. The Spatial Frame of Reference Continuum (SFoRC; Hanley et al. 2020) depicts a figure in concentric spatial zones from body-centered to environment-encompassing, indexing the degree to which the spatial sense of self extended beyond the body. The Small Self pictographic item was adapted from Vidal et al. (2025), who presented it as an ad hoc item. We included it because Vicereality is organized around a spatial metaphor and the item represents self-related experience through the spatial size of the self relative to its surroundings. All three yield a single ordinal score per condition. We recreated all three pictographs as SVG graphics for VR presentation, using the previously published versions as visual references. The vector format allowed direct and flexible control over graded visual properties, including transparency, line width, dash spacing, relative figure size, and spatial extent, while enabling sharp rendering at headset resolution without the resolution loss associated with scaling raster images. The SVG source was refined with large language model assistance. The OSF reproducibility package (https://osf.io/64mwk/) provides the recreated stimulus set, SVG source files, administered item keys, and study materials; Supplementary Section S6 provides the corresponding reproducibility map.

2.4.3 Altered States of Consciousness

3D-ASCr composites. The revised 3D-ASCr higher-order factor model became available after preregistration. It provides a parsimonious contemporary summary of the same preregistered 11D-ASC responses by organizing them into Positive, Distressing, and Perceptual Effects. This level of aggregation matches our broad theoretical interest in positively versus negatively valenced experience, while retaining a perceptual dimension, and avoids overinterpreting lower-order dimensions for which no specific predictions were made. We therefore use the 3D-ASCr composites as the main-text summary. Supplementary Section S3 reports complete analyses of all 11 preregistered 11D-ASC dimensions.

11D-ASC Questionnaire. We assessed altered states with the 11-Dimensional Altered States of Consciousness Rating Scale (11D-ASC; 42 items, 0–100 visual analogue scales), a validated lower-order factorization of the OAV/APZ framework (Studerus et al. 2010; Dittrich 1998; Hovmand et al. 2024; Prugger et al. 2022). The 11 subscales were Experience of Unity (EU), Spiritual Experience (SE), Blissful State (BS), Insightfulness (IS), Disembodiment (DIS), Impaired Control and Cognition (ICC), Anxiety (ANX), Complex Imagery (CI), Elementary Imagery (EI), Audio-Visual Synesthesia (AVS), and Changed Meaning of Percepts (CMP). Participants completed one condition-specific rating after each block.

Temporal Experience Tracers (Exploratory). Temporal experience tracers provide psychometric time-series ratings of the same 11 experiential dimensions.

For each dimension, one prompt consolidated the constituent questionnaire items into a construct-level description. After each condition, participants retrospectively drew the 11 intensity curves on a world-space time-by-intensity surface using the VR controller. Controller position was projected onto the drawing plane, input was gated through a trigger hold, and points were emitted at constant spatial spacing regardless of hand speed. Each trace was reduced to peak intensity, representing the strongest reported moment, and area under the curve (AUC), representing intensity accumulated across reconstructed session time (Figure 5). We focused on these two summaries because no hypothesis required finer temporal features such as onset, timing, slope, or variability. We treat the tracer analyses as exploratory; Supplementary Section S5 provides the complete implementation, data reduction, analyses, and results.

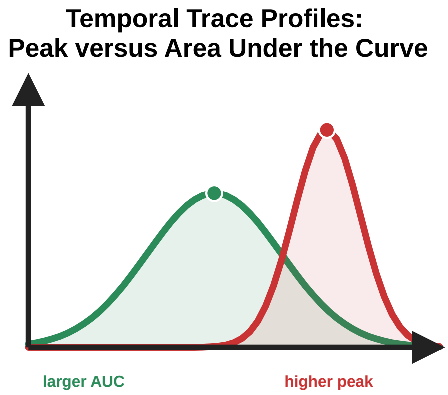

Fig. 5 Temporal experience tracers: peak and cumulative intensity. These hypothetical profiles illustrate information retained by psychometric time-series intensity ratings but not by a single retrospective visual-analogue-scale response. The red profile has a higher momentary peak, whereas the broader green profile has a larger area under the curve (AUC), reflecting greater cumulative intensity. Analyses in this paper focus on these two summaries because no hypothesis required finer-grained temporal features.

2.4.4 Blinding Questionnaire

Blinding was assessed in two phases. In the pre-unmasking phase, participants were told that the study investigated how subtle differences in visual aesthetics might influence perception in virtual reality. We constructed a 12-item ad hoc masked-comparison questionnaire for this study. For each item, participants chose whether the feature was stronger in Scenario 1, stronger in Scenario 2, equal in both scenarios, or absent in both scenarios. The chance baseline was 0.25. Scenario labels referred to the two virtual-reality conditions in encounter order. The item set covered three hidden domains. These were manipulation-relevant symmetry/coordination cues that genuinely differed between the two VR conditions, matched comparison features that were present similarly in both conditions and therefore should be judged as equal, and foil (sham) features that were not present in either condition and therefore should be judged as absent. There were four items per domain.

In shorthand, the four symmetry-related items asked whether one scene more strongly showed transient coordination and ordering of the particle field, including particles moving as a coordinated group, becoming more organized or structured at certain moments, showing wave-like coordinated motion-and-color changes, or shifting from chaotic to ordered movement during breath holds. The four neutral matched items asked about features that were present in both VR conditions to a similar degree, namely particles becoming brighter during breath holds, leaving faint motion trails during exhalation, remaining in gentle ongoing motion across the surface, or appearing to rise and fall like ocean waves. The four sham items were constructed as absent cross-modal or pulsing foils, asking about the music becoming quieter, the music seeming to slow down, particle motion and color changing in response to the music, or particles showing a size-based pulsing effect. To keep administration uniform while avoiding an obviously grouped structure, items from all three domains were intermixed in a single randomized order that was generated once in advance and then presented in that same order to all participants.

In the post-unmasking phase, participants were told that one condition featured brief visual synchronization during breath holds while the other remained irregular throughout. They indicated which session was more symmetric via a three-option forced choice (*first*, *second*, or *I don't know*), followed by a confidence rating.

2.5 Preregistration and Reproducibility

The study was preregistered on AsPredicted (#262,545). The unchanged registry record, all deidentified participant-level data, statistical analysis scripts and outputs, figure-generation scripts, and study materials were uploaded to OSF as a reproducibility package (https://doi.org/10.17605/OSF.IO/64MWK). We retained all preregistered design elements and outcome families. The preregistered analysis plan specified a repeated-measures ANOVA with condition as the within-subjects factor and presentation order as a covariate for each outcome, two planned contrasts (pooled VR versus dark-screen control, and symmetric versus asymmetric), and parallel Bayesian analyses. Adherence was to be quantified as phase-duration error with a cycle-validity exclusion rule, as described and motivated in Section 2.4.1. We deviated from this plan in several respects. Supplementary Section S1 summarizes the deviations from preregistration and their rationale.

Adherence metric. Phase-duration error was replaced by PLV because the recorded phase-label data were too fragmented to yield reliable per-phase error estimates without introducing arbitrary binning decisions (see Section 2.4.1 for a full account). The associated exclusion rule was changed from 70%-valid-cycle plus 2-SD mean-error to a lower-tail global IQR rule on condition-level PLV values, because PLV does not decompose into per-cycle validity counts.

Omnibus condition test. The preregistered RM-ANOVA with condition and order was replaced by an order-adjusted mixed-effects likelihood-ratio test comparing a model with condition and block-position order against an order-only model, with participant as random intercept. The realized counterbalanced design was more naturally represented by condition-specific block-position covariates than by a single omnibus order term, and this formulation preserves the intended within-subject and order-adjusted comparison without forcing an approximation to the realized order structure. A condition-only repeated-measures ANOVA without the order covariate is retained in the OSF outputs as a sensitivity check.

Planned contrasts. The two preregistered planned contrasts were implemented as standalone paired t-tests, which yield the same paired contrast statistics as ANOVA follow-ups in this within-subject setting. These serve as the confirmatory anchor for all multi-test outcome families in the Results section, and for significant planned contrasts in the altered-state families we additionally report paired common-language effect sizes (CLES), defined as the proportion of participants whose score was higher in the VR condition than in the dark-screen control (McGraw and Wong 1992).

Robustness and multiplicity. Because Shapiro–Wilk checks indicated non-normality for several condition-level and difference-score distributions, Wilcoxon signed-rank tests and Friedman omnibus tests were added as distribution-free robustness checks. The preregistration did not specify a multiple-comparison correction strategy. We applied Benjamini–Hochberg false-discovery-rate (FDR) correction within each multi-test outcome family (11 ASC dimensions, 3 derived 3D-ASCr dimensions, 3 self-related variables). Main-text confirmatory interpretations are anchored in the planned contrasts with uncorrected p values. FDR-adjusted q values and the full robustness-check output are reported in the OSF repository and are mentioned in the main text only when they materially qualify the confirmatory pattern.

3D-ASCr composites. The revised 3D-ASCr scoring scheme (Stocker et al. 2025) became available after preregistration. Because these composites are deterministic averages of the preregistered 11D-ASC subscale scores, they do not introduce additional analytic degrees of freedom and are presented as a dimensionality-reduced summary rather than as separate confirmatory endpoints (see Section 2.4.3).

Bayesian analyses. Bayesian analyses were preregistered and run in parallel on the same included sample ($N = 39$). Omnibus Bayes factors compared a point-null model without condition (H_0: outcome $\sim$ order + participant random intercept) against a model that additionally included condition (H_1: outcome $\sim$ condition + order + participant random intercept), using the BayesFactor R package with explicit prior settings (fixed-effect $r = 0.50$, continuous-covariate $r = \sqrt{2}/4 \approx 0.354$, random-effect nuisance $r = 1.00$; Morey and Rouder 2023; van Doorn et al. 2021; Kruschke 2021). For the two planned paired contrasts, we compared H_0 ($\delta = 0$) against two-sided H_1 ($\delta \neq 0$) using a Bayesian paired t-test with the default Cauchy prior on standardized effect size ($r = \sqrt{2}/2 \approx 0.707$; Morey and Rouder 2023; van Doorn et al. 2021). Preregistered one-sided ordered checks (symmetric > asymmetric > dark-screen control) were restricted to EU, BS, and AVS. We report BF_{10} values (and $\mathrm{BF}_{01} = 1/\mathrm{BF}_{10}$ when the data favor the null) as relative evidence rather than as posterior probabilities or effect sizes (van Doorn et al. 2021; Kruschke 2021). Posterior median standardized effect sizes with 95% credible intervals and prior-robustness checks are reported in the OSF repository.

Blinding analyses. Pre-unmasking accuracy was scored using the documented answer key for the four symmetry, four neutral, and four sham items and across all 12 items. Mean accuracy was compared with the 0.25 chance baseline using Wilcoxon signed-rank tests, with Benjamini–Hochberg FDR correction across the four comparisons. Post-unmasking guesses were classified as correct (R), incorrect

(W), or uncertain (U) from the participant's selected active-VR session and condition order. The standard single-group Bang blinding index was calculated as BBI $= (R - W)/(R + W + U)$ (Bang et al. 2004; Kolahi et al. 2009), with a participant-bootstrap 95% confidence interval. Effective blinding was considered established only if the entire interval fell within $[-0.2, 0.2]$ (Qin et al. 2024; Kolahi et al. 2009). We also report an exploratory confidence-weighted directional index, calculated as the confidence-weighted proportion of correct directional guesses minus 0.5. The standard Bang interpretation bands were not applied to this modified index. To test whether pre-unmasking symmetry accuracy predicted subsequent condition identification, we calculated a point-biserial correlation between the 0–4 symmetry score and final-guess correctness, classified as correct versus incorrect or uncertain, with an exact two-sided permutation p value.

Exploratory analyses. We fit condition $\times$ PLV interaction models to test whether breathing synchronization moderated subjective effects. We also examined whether VR-related changes in the pictographic measures covaried with changes in ASC outcomes. The primary estimand was the Spearman correlation between pooled-VR-minus-dark-screen-control changes in each pictograph and the corresponding ASC outcome. Supplementary Section S4 reports the complete multiplicity-corrected analysis, including symmetric-minus-asymmetric change correlations and order-adjusted participant-aware condition-interaction models.

3 Results

Across the four recurring outcome families, condition effects were consistently driven by the contrast between virtual-reality conditions and the dark-screen control. No outcome showed a reliable symmetric-versus-asymmetric difference. We report the results in the same family order used in Methods: adherence, self-related measures, altered states of consciousness, and the blinding questionnaire. For all multi-test families, confirmatory interpretations are anchored in the preregistered planned contrasts evaluated as paired t-tests, with the order-adjusted mixed-effects omnibus noted for each family as context. Wilcoxon signed-rank robustness checks, Friedman omnibus tests, and FDR-adjusted q values are documented in the OSF repository and are mentioned here only when they materially qualify the paired t pattern.

3.1 Sample and Exclusions

The dataset contained records from all 47 tested participants. Condition-level PLV values from all participants were used to apply the quality-screening procedure described in Section 2.4.1 and illustrated in Figure 6. This procedure excluded 8 participants and yielded a final analytic sample of $N = 39$. The 39 included participants each contributed a complete within-subject triad and constituted the final sample used for all inferential analyses.

3.2 Outcome Families

3.2.1 Adherence (PLV)

To test whether visual biofeedback improved adherence to the guided box-breathing rhythm, we compared PLV across conditions. Mean PLV was $M = 0.787$ ($SD = 0.141$) for the dark-screen control, $M = 0.836$ ($SD = 0.127$) for symmetric VR, and $M = 0.860$ ($SD = 0.094$) for asymmetric VR. The planned pooled-VR-versus-dark-screen-control contrast confirmed this difference, $t(38) = 3.47$, $p = .001$, $d_z = 0.56$, 95% CI for the mean PLV difference $[0.026, 0.097]$. The symmetric-versus-asymmetric contrast was not significant, $t(38) = -1.13$, $p = .267$, $d_z = -0.18$. The order-adjusted mixed-effects omnibus supported a condition effect, $\chi^2(2) = 12.66$, $p = .002$. The corresponding Bayesian planned contrasts showed that the pooled-VR-versus-dark-screen-control difference was about 24.1 times more likely than no difference ($\mathrm{BF}_{10} = 24.11$), whereas the symmetric-versus-asymmetric data were about 3.2 times more likely under a no-difference model ($\mathrm{BF}_{01} = 3.22$).

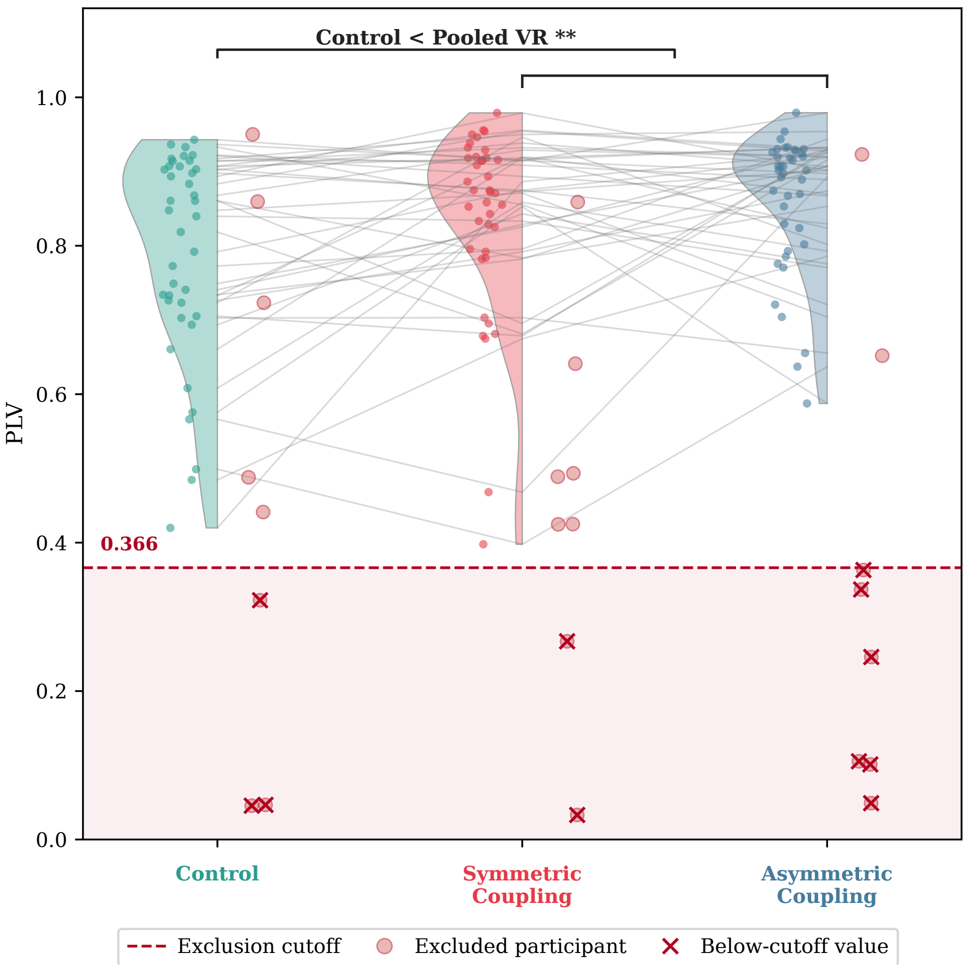

Fig. 6 Controller-derived feedback synchrony (PLV) by condition for the selected sample ($N = 39$), with condition-wise distributions and paired individual trajectories in gray. The dashed red line marks the pooled lower-tail IQR exclusion cutoff (0.366); hollow circles and red crosses indicate flagged participants and their below-cutoff values, respectively. If any single condition fell below the cutoff, the participant was excluded entirely from all analyses across all conditions.

3.2.2 Self-Related Measures

To assess whether the breath-responsive virtual environment shifted subjective self-representation, we compared three pictographic measures across conditions. Only the spatial frame-of-reference continuum showed a condition effect. Participants reported a more spatially extended sense of self during virtual-reality conditions, with $M = 3.15$ ($SD = 1.41$) for the dark-screen control, $M = 4.18$ ($SD = 1.82$) for symmetric VR, and $M = 4.26$ ($SD = 1.73$) for asymmetric VR. The planned pooled-VR-versus-dark-screen-control contrast confirmed the effect, $t(38) = 4.28$, $p < .001$, $d_z = 0.69$, 95% CI $[0.56, 1.57]$. The symmetric-versus-asymmetric contrast was not significant,

$t(38) = -0.32$, $p = .752$. The order-adjusted mixed-effects omnibus converged with this pattern, $\chi^2(2) = 20.04$, $p < .001$. Neither perceived body-boundary salience ($t(38) = -0.17$, $p = .864$) nor small self ($t(38) = 1.39$, $p = .173$) showed a significant VR-versus-dark-screen-control contrast, and no self-related measure showed a significant symmetric-versus-asymmetric contrast.

Bayesian comparisons agreed with this pattern at the planned-contrast level. For spatial frame of reference, the observed data were approximately 204 times more likely under a pooled-VR-versus-dark-screen-control difference model than under the no-difference model ($\mathrm{BF}_{10} = 204.44$). Perceived body boundaries favored the no-difference model ($\mathrm{BF}_{01} = 5.71$), whereas the Small Self result provided only weak and inconclusive evidence in the null direction ($\mathrm{BF}_{01} = 2.39$).

Exploratory change-score correlations tested whether VR-related changes in the pictographic measures covaried with ASC dimensions selected for their thematic relevance to spatial and bodily self-experience; Supplementary Section S4 reports the complete multiplicity-corrected screen. Greater SFoRC extension covaried strongly with Experience of Unity ($\rho = .68$, $q = .002$) and more moderately with Disembodiment ($\rho = .36$, $q = .081$). Greater Small Self ratings covaried with Blissful State ($\rho = .52$, $q = .012$) and Spiritual Experience ($\rho = .46$, $q = .021$), whereas boundary dissolution did not covary reliably with Disembodiment ($\rho = .11$, $q = .591$). No symmetric-minus-asymmetric change correlation survived correction. The supported associations therefore concern pooled VR-related changes rather than differential correspondence with the two visual mappings.

3.2.3 Altered States of Consciousness

We report the altered-state family using the 3D-ASCr composites as a dimensionality-reduced overview and the underlying 11D-ASC subscales. The clearest pattern is stronger positive and perceptual VR-related effects, with a weaker distress-related component and no reliable symmetric-versus-asymmetric difference.

3D-ASCr composites.

Summarizing the 11D-ASC data through the 3D-ASCr composites (see Section 2.4.3 for derivation), the three higher-order dimensions showed the following pattern. Perceptual Effects had $M = 11.41$ ($SD = 12.61$) in the dark-screen control, $M = 27.25$ ($SD = 19.92$) in symmetric VR, and $M = 25.59$ ($SD = 17.24$) in asymmetric VR. Positive Effects had $M = 18.88$ ($SD = 15.39$) in the dark-screen control, $M = 25.64$ ($SD = 19.48$) in symmetric VR, and $M = 26.46$ ($SD = 17.84$) in asymmetric VR. Distressing Effects had $M = 12.33$ ($SD = 10.26$) in the dark-screen control, $M = 17.33$ ($SD = 12.66$) in symmetric VR, and $M = 17.02$ ($SD = 13.18$) in asymmetric VR. All three composites showed significant pooled-VR-versus-dark-screen-control planned contrasts: Perceptual Effects $t(38) = 4.90$, $p < .001$, $d_z = 0.79$; Positive Effects $t(38) = 3.22$, $p = .003$, $d_z = 0.51$; Distressing Effects $t(38) = 2.56$, $p = .014$, $d_z = 0.41$. Expressed on the native 0–100 scale, pooled VR increased Perceptual Effects by 15.0 points, Positive Effects by 7.2 points, and Distressing Effects by 4.8 points. These corresponded to a large effect for Perceptual Effects and moderate effects for Positive and Distressing Effects, with 76.9%, 69.2%, and 76.9% of participants, respectively, scoring higher in VR than in the dark-screen control.

No composite showed a symmetric-versus-asymmetric difference (all $p \geq .357$). Order-adjusted mixed-effects omnibus tests converged for all three composites (Perceptual Effects $\chi^2(2) = 32.83$, Positive Effects $\chi^2(2) = 14.32$, Distressing Effects $\chi^2(2) = 9.11$; all $p \leq .011$). Bayesian omnibus comparisons favored condition models for all three composites, with the strongest evidence for Perceptual Effects ($BF_{10} = 121{,}810$), followed by Positive Effects ($BF_{10} = 30.49$) and Distressing Effects ($BF_{10} = 3.45$; Figure 7).

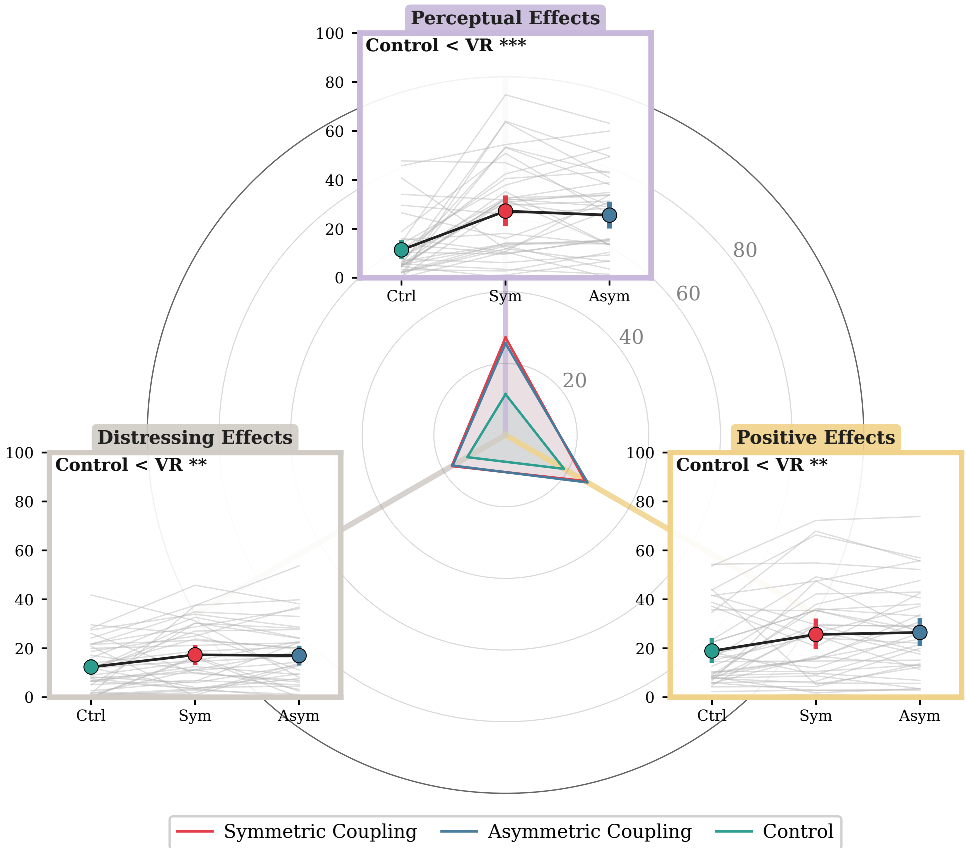

Fig. 7 Condition-wise distributions for the three 3D-ASCr composites (N = 39). All three dimensions increased in VR relative to the dark-screen control, with the largest effect for Perceptual Effects; no symmetric-versus-asymmetric differences emerged.

11D-ASC questionnaire.

Six subscales showed significant pooled-VR-versus-dark-screen-control planned contrasts: Experience of Unity ($t(38) = 3.50$, $p = .001$, $d_z = 0.56$), Spiritual Experience ($t(38) = 2.58$, $p = .014$, $d_z = 0.41$), Disembodiment ($t(38) = 2.59$, $p = .014$,

$d_z = 0.41$), Elementary Imagery ($t(38) = 4.71$, $p < .001$, $d_z = 0.75$), Audio-Visual Synesthesia ($t(38) = 5.15$, $p < .001$, $d_z = 0.82$), and Changed Meaning of Percepts ($t(38) = 3.30$, $p = .002$, $d_z = 0.53$). On the native ASC scale, these pooled-VR increases ranged from 6.6 points for Spiritual Experience to 25.6 points for Audio-Visual Synesthesia. Standardized effects ranged from moderate ($d_z = 0.41$ for Spiritual Experience and Disembodiment) to large ($d_z = 0.82$ for Audio-Visual Synesthesia), with 67.9% to 87.2% of participants scoring higher in VR than in the dark-screen control across these six dimensions. Impaired Control and Cognition was significant at the uncorrected level ($t(38) = 2.15$, $p = .038$, $d_z = 0.34$) but borderline after within-family FDR adjustment ($q = .060$). Anxiety did not reach significance ($t(38) = 1.92$, $p = .063$). No subscale showed a significant symmetric-versus-asymmetric contrast (all $p \geq .166$). Condition-wise descriptives for all 11 subscales are reported in the OSF repository.

Order-adjusted mixed-effects omnibus tests converged with the planned-contrast pattern for most subscales, with one divergence. Impaired Control and Cognition showed a marginal mixed-model omnibus ($p = .040$, $q = .062$) but was supported by a non-parametric Friedman robustness check ($p < .001$); this divergence is documented in the OSF repository.

Bayesian planned contrasts supported the same six dimensions. The strongest evidence concentrated in Audio-Visual Synesthesia ($BF_{10} = 2{,}408$) and Elementary Imagery ($BF_{10} = 670$), followed by Experience of Unity ($BF_{10} = 64.0$), Changed Meaning of Percepts ($BF_{10} = 12.6$), Spiritual Experience ($BF_{10} = 5.60$), and Disembodiment ($BF_{10} = 4.94$). Blissful State, Insightfulness, and Complex Imagery favored the null (BF_{01} range 3.3–8.8), while Impaired Control and Cognition and Anxiety remained inconclusive. Symmetric-versus-asymmetric Bayesian contrasts uniformly favored no difference (BF_{01} range 2.3–5.7; per-dimension values in the OSF repository).

The preregistered one-sided ordering checks were more selective. For the predicted asymmetric-greater-than-dark-screen-control step, the data supported the directional effect for Experience of Unity ($BF_{10} = 87.7$) and Audio-Visual Synesthesia ($BF_{10} = 745.0$) but not for Blissful State ($BF_{01} = 2.68$). For the predicted symmetric-greater-than-asymmetric step, the data favored the null for Experience of Unity and Blissful State ($BF_{01} = 8.07$ and 12.43) and remained inconclusive for Audio-Visual Synesthesia ($BF_{10} = 0.79$).

For descriptive cross-study context, we compared the pooled Viscereality VR profile with four published breathwork datasets that reported compatible ASC summaries: conscious connected breathing (Bahi et al. 2024a), facilitated circular breathwork (Havenith et al. 2025b), high-ventilation breathwork accompanied by music in an MRI/laboratory study (Kartar et al. 2025), and Holotropic Breathwork (Uthaug et al. 2021). Viscereality was lower than Bahi et al. across all three 3D-ASCr composites; it was close to Havenith et al. on Positive and Perceptual Effects but lower on Distressing Effects. Relative to Kartar et al., it showed lower Positive Effects but higher Distressing and Perceptual Effects, and it was higher than Uthaug et al. across all three composites in the extracted summary. These overlays compare raw profile means rather than matched effect sizes and confound breathing technique, session duration, participant experience, music and facilitation, and control structure. They are therefore descriptive only (Figure 9).

3.2.4 Blinding Questionnaire

Pre-unmasking responses showed selective but incomplete sensitivity to the symmetry manipulation. Mean symmetry-item accuracy was 44.2%, equivalent to approximately 1.77 of four items, and exceeded the 25% chance baseline (FDR-adjusted $q < .001$). Neutral-item accuracy was 27.6% ($q = .610$), and sham-item accuracy was 32.7% ($q = .100$); neither was clearly above chance. Across all 12 items, mean accuracy was 34.8% ($q < .001$). Participants therefore detected some genuine symmetry features, but performance remained well below perfect identification.

Post-unmasking guesses were nearly balanced: 17 participants guessed correctly (43.6%), 19 guessed incorrectly (48.7%), and 3 were uncertain (7.7%). The standard Bang blinding index was close to zero (BBI $= -.051$, 95% bootstrap CI $[-.359, .256]$). Confidence weighting changed little: the exploratory confidence-weighted index was $-.016$ (95% bootstrap CI $[-.203, .168]$, $n = 35$ directional responses). Both intervals crossed zero, indicating no reliable tendency toward either correct or opposite identification. However, the standard BBI interval extended beyond the predefined $[-0.2, 0.2]$ blinding region. The results were therefore consistent with effective blinding but too uncertain to rule out meaningful identification in either direction.

To test whether pre-unmasking sensitivity predicted subsequent condition identification, we related the 0–4 symmetry score to final-guess correctness, classified as correct versus incorrect or uncertain. Pre-unmasking symmetry-detection accuracy was descriptively higher among participants who subsequently identified the correct condition (2.00 versus 1.59 of four items), but the association was small and inconclusive (point-biserial $r = .18$, exact two-sided permutation $p = .33$; Figure 8d). Thus, better masked detection did not reliably predict the final condition guess in this sample.

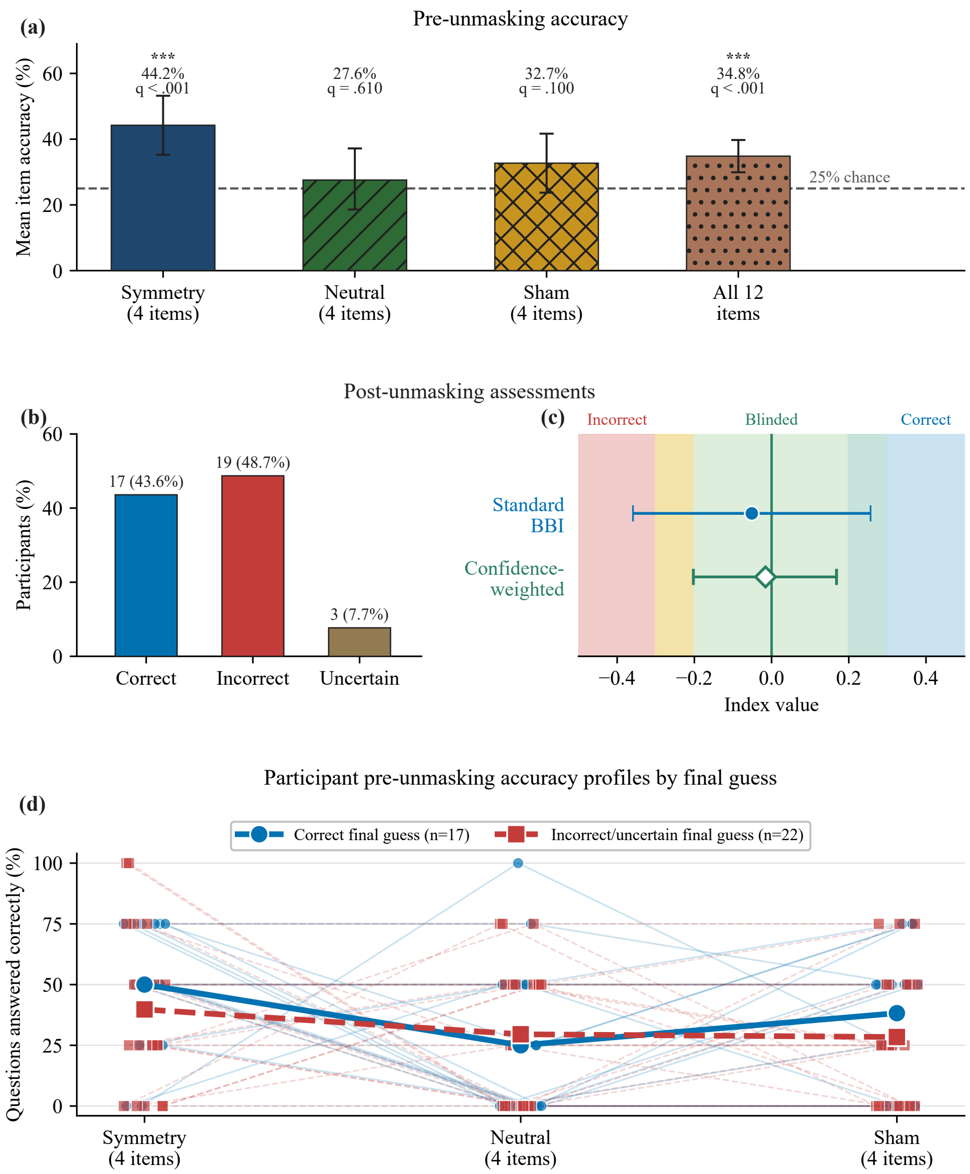

Fig. 8 Blinding and manipulation-awareness summary ($N = 39$). (a) Mean pre-unmasking accuracy for the four symmetry, four neutral, and four sham items and across all 12 items. The dashed line marks the 25% chance baseline; asterisks denote FDR-adjusted $q < .001$. (b) Post-unmasking guesses: 17 correct, 19 incorrect, and 3 uncertain. (c) Standard Bang blinding index (filled blue circle) and exploratory confidence-weighted directional index (open green diamond), with participant-bootstrap 95% confidence intervals. The shaded regions indicate incorrect, blinded, and correct identification and apply only to the standard Bang index. (d) Participant-level pre-unmasking accuracy across the three four-item categories, grouped by final-guess correctness. Thin lines and small markers represent participants; heavier lines and larger markers represent group means. Incorrect and uncertain guesses are combined.

3.3 Exploratory Checks

Exploratory mixed-model checks for order and PLV moderation are reported in the OSF repository. No FDR-corrected order main effects, condition × order, condition × PLV, or condition × order × PLV interactions emerged for any ASC, 3D-ASCr, or self-related outcome family. For PLV itself, there was a main effect of order, $\chi^2(1) = 8.28$, $p = .004$, but no condition × order interaction, $\chi^2(2) = 0.37$, $p = .833$, indicating that block order did not change the between-condition adherence pattern. A planned linear contrast across session position was significant, $t(38) = -3.16$, $p = .003$, 95% CI $[-0.100, -0.022]$, indicating that PLV decreased rather than increased across sessions and therefore aligning more with waning engagement or fatigue than with a practice-improves account.

4 Discussion

4.1 Summary of findings

This study provided a first controlled empirical evaluation of Viscereality with naive participants. Relative to audio-guided box breathing in the dark-screen control, the two VR conditions improved breathing adherence, shifted spatial self-experience toward a more expanded frame of reference, and produced an altered-state profile in which perceptual and positive effects outweighed mild distressing effects, with no adverse events across 47 tested participants. However, the preregistered aesthetic manipulation that varied the symmetry of the particle dynamics did not reliably differentiate the two active VR conditions. Taken together, these findings indicate that the paradigm produced a structured experiential profile during a brief breathing session with naive participants, but stronger control comparisons and objective measures are needed to determine the specificity and depth of these effects.

4.2 Breathing adherence

Participants in both active VR conditions followed the instructed breathing pattern more closely than in the dark-screen control. The display responded to participants' ongoing breathing rather than to the audio guide, so it made respiration continuously visible without telling participants how well they were matching the instructed pacing. One plausible interpretation is that the breath-responsive environment helped adherence by keeping participants engaged and by giving them the sense that their breathing was producing the visual experience. Previous findings are consistent with this. Breath-coupled VR feedback has been linked to improved focus on breathing and greater diaphragmatic movement (Blum et al. 2020; Rockstroh et al. 2021), greater focused attention and less distraction than non-VR biofeedback (Rockstroh et al. 2019), and increased interoceptive sensibility (Goral et al. 2024a). A simpler explanation also remains possible. Guided breathing in the dark-screen control is presumably less engaging than the same task in an immersive environment, and the adherence metric itself was derived from how closely breathing matched the trajectory expressed by the visual display. The study would have benefited from a more stringent control, such as a visually matched scene that does not respond to breathing or one that replays a prerecorded animation, because the engaging visual environment on its own may have contributed to the effect. This caveat matters because controlled comparisons in the VR breathing literature have not consistently shown VR-based interventions to outperform matched non-VR approaches on clinical outcomes (Cortez-Vázquez et al. 2024; Pancini et al. 2025). The present adherence finding should therefore be treated as preliminary until tested against a more stringent control.

4.3 Self-related experience

Of the three self-related measures, only the spatial frame-of-reference continuum showed a condition effect. We had reason to expect body-boundary salience to shift as well because prior work reports reduced perceived boundary salience during meditation (Dambrun 2016); in one study that administered both scales, mindfulness training reduced perceived body boundaries and expanded allocentric spatial framing in tandem, with the boundary change mediating the spatial shift (Hanley et al. 2020). That pattern did not emerge here, perhaps because the two constructs respond to partly different triggers. Body-boundary dissolution appears to shift most reliably during practices that direct sustained attention toward the body itself, such as body-scan meditation (Dambrun 2016), mindfulness training (Hanley et al. 2020), and floatation-REST (Hruby et al. 2024). Spatial frame of reference, by contrast, has been linked not only to body-directed contemplative practice (Hanley et al. 2020) but also to perceived expansion of the physical frame of reference beyond the body (Dambrun 2023). Because Viscereality acts on the surrounding visual-spatial environment rather than directing attention toward the body, it may have engaged the spatial-frame construct through this environmental route without producing the body-focused conditions under which boundary dissolution typically occurs. Of the three scales, spatial frame of reference is also the one most closely aligned with the broader question motivating this research programme, namely whether breath-coupled changes in surrounding space can alter the felt extension of self into the environment. Previous work has shown that mapping respiration onto an externalized virtual avatar can modulate body ownership (Monti et al. 2020b) and self-location (Adler et al. 2014; Betka et al. 2020) through afferent respiratory pathways (Betka et al. 2020). However, this result requires cautious interpretation. The concentric depiction of the SFoRC scale may have resembled the expanding and contracting particle sphere shown in VR, which raises the possibility that participants selected more spatially expansive depictions because the scale imagery matched what they had just seen rather than because their felt self-boundary had shifted. This demand-characteristics concern cannot be ruled out with pictographic self-report alone and represents a clear limitation of the present measurement approach. The finding should accordingly be treated as preliminary pending corroboration from measures that do not share visual features with the stimulus, such as peripersonal space tasks. If confirmed by such measures, the spatial-frame shift would support continued investigation of whether mapping respiration onto surrounding space can modulate the felt extension of self into the environment.

4.4 Altered states of consciousness

Within the context of a 10-minute box-breathing exercise with ambient background music, *Viscereality* increased all three higher-order dimensions of the Altered States of Consciousness Rating Scale relative to the dark-screen control, which provided only a central fixation marker and minimal visual breathing-performance feedback. Perceptual Effects showed the largest increase, Positive Effects a moderate increase, and Distressing Effects a smaller increase. None differed reliably between the two VR mappings that manipulated visual symmetry, and participants showed only limited feature-level sensitivity to this manipulation without reliable condition-level identification.

The fine-grained subscale decomposition and item-content analysis reveal a more specific experiential profile (Supplementary Section S3.4). The perceptual shift primarily concerned simple patterns, colors, lights, and flashes, together with impressions that sound changed visual shapes or colors. These elementary visual phenomena closely matched features supplied directly by the particle environment, so their increase partly reflects overlap between the intervention and the questionnaire criteria. Participants also attributed visual changes to sound even though the same background music was presented in every condition and did not control the display. One plausible interpretation is that the richer and more dynamic visual stream provided more perceptual material that could be bound to, or retrospectively attributed to, concurrent sound, rather than producing audio-visual synaesthesia in the stricter sense. A similar distinction has been raised in psychedelic research, where LSD increased spontaneous synaesthesia-like reports that lacked the consistency and inducer specificity associated with genuine synaesthesia (Terhune et al. 2016). The Positive Effects increase was concentrated in items reflecting a loosening of separation between self, surroundings, and time, heightened significance and emotional salience of ordinary objects, sensations of floating or diminished bodily presence, and moments of awe. The smaller Distressing Effects increase had the character of diffuse destabilization, encompassing reduced agency, difficulty organizing thought, isolation, unexplained apprehension, and an altered or threatening quality of the surroundings. Neither constituent distress-related subscale remained reliable after correction across the fine-grained analyses. We therefore interpret the composite increase as an accumulation of mild and compatible unpleasant experiences rather than as a pronounced or clearly localized adverse response.

To interpret the subjective effects beyond their measurement caveats, it helps to situate them relative to similar induction contexts. The 11D-ASC was designed to enable comparisons across different techniques and substances (Dittrich 1998; Studerus et al. 2010), and a growing body of cross-study data now supports such contextualization (Prugger et al. 2022). As shown in Figure 9, we compared the present profile with four breathwork datasets: conscious connected breathing (Bahi et al. 2024b), facilitated circular breathwork (Havenith et al. 2025b), high-ventilation breathwork with music in an MRI/laboratory setting (Kartar et al. 2025), and Holotropic Breathwork (Uthaug et al. 2021). The marked variation among the profiles displayed in the figure indicates that subjective-effect ratings depend substantially on the breathing technique and the wider context in which it is practised. Differences in session duration, participant experience, music, facilitation, setting, and study design preclude direct comparison, so the figure provides qualitative context rather than evidence of equivalence. More meaningful comparisons require understanding how breathing technique and context shape individual 11D-ASC subscales, and which subscales are most context-dependent. Future studies should combine subscale-level analyses with relevant neurophysiological biomarkers to distinguish shared physiological mechanisms from intervention-specific and contextual effects across breathwork with and without VR.

The comparison in Figure 9 nevertheless highlights an important practical distinction. The reference studies predominantly involved high-ventilation or circular-breathing techniques conducted over longer periods, often with music and facilitator support. Such protocols may be unsuitable for some individuals vulnerable to panic, syncope, hypocapnia, or cardiovascular complications (Fincham et al. 2023), while their duration and facilitation requirements may limit integration into time-constrained clinical settings. By contrast, the present intervention lasted only 10 minutes and produced pronounced Perceptual Effects that fell within the range of some longer breathwork studies, while Distressing Effects remained comparatively low and no adverse events were observed in the present sample. These findings do not establish clinical safety or efficacy, but they identify brief bioresponsive VR paired with slow-paced breathing as a candidate for further study in settings where relaxation-oriented breathing exercises are already established, including neurorehabilitation. Research in clinical populations should determine whether the visual feedback adds meaningful benefit to conventional breathing practice and should directly assess feasibility, accessibility, tolerability, and safety.

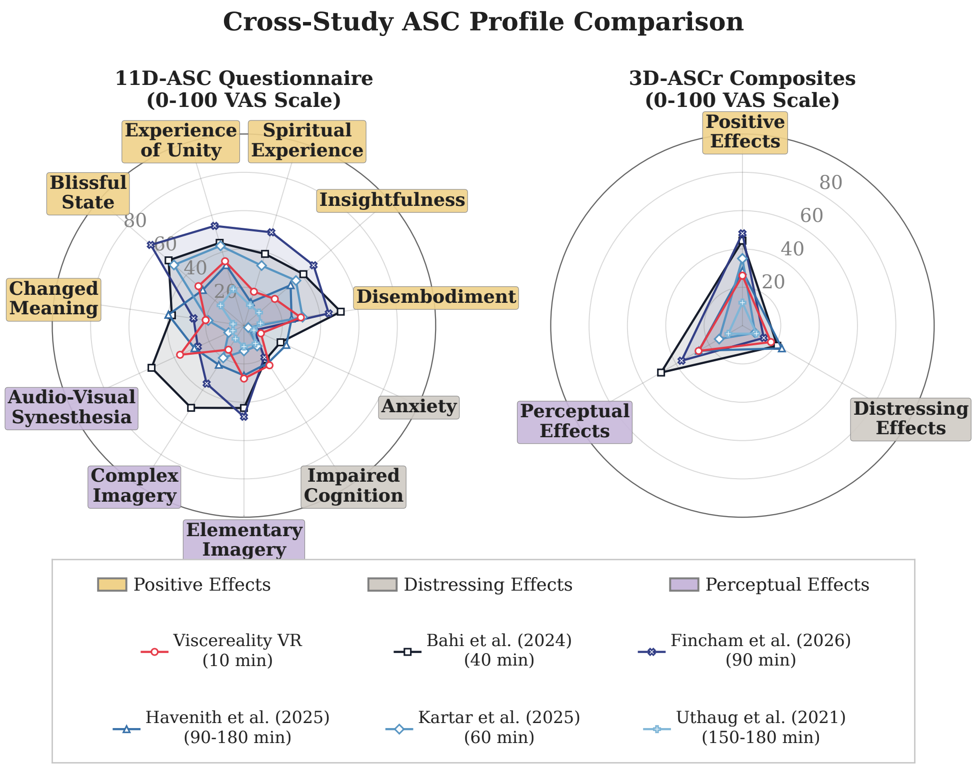

Fig. 9 Descriptive cross-study comparison of altered-state profiles from five named datasets. The left panel shows 11D-ASC subscale means and the right panel shows 3D-ASCr composite means. The Viscereality profile pools the symmetric and asymmetric VR conditions. Lines show raw 0–100 profile means, not standardized effect sizes; differences in technique, duration, sample, music, facilitation, and study design preclude inferential comparison. Legend order is purely graphical and does not imply a numbered study sequence.

Looking closer at the differences across subscales, our VR intervention produced somewhat higher scores on Experience of Unity, Spiritual Experience, Blissful State, and Disembodiment, while circular breathwork produced roughly twice the intensity on Insightfulness and Changed Meaning of Percepts. Bayesian analyses confirmed that these latter two dimensions were largely unaffected by the addition of VR relative to the dark-screen control. One qualitative difference that may partially account for this pattern is where visual awareness is directed. Traditional breathwork typically involves eye shades (Havenith et al. 2025b; Grof 2019) and frames the experience as an inward-directed process, with attention oriented toward bodily sensation (Grof 2019; Rhinewine and Williams 2007). Our VR paradigm directs visual awareness outward, toward a surrounding environment that responds to the breath. The somewhat higher unity and disembodiment scores in our VR intervention may in part reflect this difference, though the many other ways the two paradigms differ make it difficult to say with certainty. Insightfulness and Changed Meaning of Percepts, where circular breathwork produced roughly twice the intensity, capture reflective and integrative processes. In psychedelic research, more challenging experiences have been found to be associated with greater retrospectively attributed meaningfulness and personal significance (Barrett et al. 2016), and it is plausible that a similar dynamic operates in breathwork, where working through a prolonged and at times distressing session gives rise to a sense of having gained something from the experience. Such effects would be unrealistic to expect after 10 minutes of relaxation-oriented breathing, nor was this the design intention. The present system was designed as a brief, pleasant, self-contained experience for augmenting breath, and in our view we achieved that.

4.5 Blinding and the null symmetry result

The blinding results distinguish masked feature-level sensitivity from post-unmasking condition identification. Before unmasking, symmetry-item accuracy exceeded chance, while neutral and sham accuracy did not. Participants therefore detected some genuine symmetry features, but only imperfectly. After unmasking, correct and incorrect guesses were nearly balanced, and both blinding indices were close to zero. Correct guessers had descriptively higher symmetry accuracy than incorrect or uncertain guessers, but the association was small and inconclusive (r = .18, exact p = .33). Overall, the results were consistent with effective blinding, but the wide standard BBI interval prevented a conclusive determination.

The fact that participants partially detected the symmetry manipulation at the feature level yet showed no net correct identification at the condition level may also point toward a potential explanation for why no outcome family showed a meaningful difference between the symmetry and anti-symmetry conditions. The geometric coherence unique to the symmetry condition emerged only during breath-hold phases, which nominally occupy half the breathing cycle. But the primary task demand was to drive the expansion and contraction of the particle sphere through inhaling and exhaling, and participants' attention was likely oriented toward this respiratory coupling rather than toward the aesthetic character of the holds. The breath holds remained visually active, as the particle field continued its turbulent oscillations throughout, but these were not the phases where participants felt they were performing the task. The effective attentional exposure to the symmetry difference was therefore considerably less than the nominal 50% of session time. The blinding data compound this ambiguity. The near-zero point estimate could reflect successful blinding, but its wide interval precludes a definitive conclusion; it could also reflect that the manipulation was too subtle or too brief to register as an experientially meaningful difference. The same perceptual subtlety that enables within-modality blinding may also prevent the aesthetic variation from producing measurable shifts in subjective experience. The present design does not provide precise control over the timing, duration, or attentional allocation of exposure to the aesthetic manipulation, and the degree to which task demands distracted from the hold-phase aesthetics is an uncontrolled variable.

Several explanations for the null symmetry result should be weighed together. The manipulation may have been too subtle to produce experiential differences at the level measured by the present outcome scales. The effective exposure may have been too brief, given that the symmetry difference was confined to breath-hold phases and that attention was likely oriented toward the respiratory coupling during inhale and exhale phases. The particular features that were varied, specifically the coupling strength governing spatial coherence, may not be the features that matter most for subjective experience in this context. Or the aesthetic difference may genuinely not produce measurable experiential shifts at this exposure level regardless of attention. The present data do not distinguish between these accounts, and doing so would require designs that isolate exposure duration, attentional allocation, and aesthetic feature type as independent variables.

There are converging indications that the types of features varied in the present system do contribute to affective processing below the threshold of explicit awareness, which suggests the null result reflects the conditions of exposure rather than the irrelevance of the features themselves. Visual symmetry carries implicit positive hedonic valence that dissociates from explicit preference ratings (Ruta et al. 2021), and the sustained posterior negativity generated by visual regularity scales with the degree of symmetry regardless of whether participants attend to it (Makin et al. 2023). Spatial features of surrounding environments, including contour geometry and perceived enclosure, activate reward-related and valence-processing circuitry during both active aesthetic judgment and passive viewing (Vartanian et al. 2013, 2024; Vessel and Rubin 2021). Interoceptive signals, including respiratory and cardiac rhythms, modulate body representation and self-related experience through largely implicit processes (Tsakiris et al. 2011b), and spatial perception engages kinesthetic and proprioceptive dimensions that extend beyond explicit evaluation (Kwon and Iedema 2022). The question for future work is therefore not whether these features matter in principle, but under what conditions of exposure, attention, and duration they produce measurable shifts in subjective experience within immersive environments.

4.6 Limitations

To probe the efficacy of our VR system as a novel altered-state induction method, a psychometric measurement scale less reliant on visual descriptors would have been more conducive to evaluating this capacity, although similar critiques exist around other instruments such as the Mystical Experience Questionnaire, which has been criticized for reliance on a religious framing (Sanders and Zijlmans 2021). To confirm whether the questionnaires and pictographic scales capture real differences reflective of the constructs they are aimed to measure, namely the system's capacity to induce altered states of consciousness and extend subjective self-boundaries, further evidence through objective markers is needed, preferably ones less dependent on the visual modality to remain orthogonal to the manipulation. Furthermore, evaluating the system's aesthetic parameters requires more reliable experimental controls, for instance over exposure time and attentional allocation to these features, and more fine-grained differentiation of what the manipulations are intended to assess. The study was conducted on Western first-year psychology students, hence the utility of our VR paradigm, for instance within clinical contexts, is limited pending further testing on tolerability in the wider population.

4.7 Future directions

The Viscereality system was designed with the aim of inducing experiences that enhance the plasticity of self-representation and dissolve the felt boundary between self and environment, which we situate within the broader phenomenology of altered states of consciousness. The spatial frame of reference finding lends preliminary support for the system's capacity to shift this boundary, but as with any retrospective self-report measure, it requires corroboration from objective measures that do not depend on the participant's post-hoc interpretation of their experience. Our future work will use peripersonal space tasks as the primary objective outcome measure. Peripersonal space indexes the spatial boundary surrounding the self through multisensory integration processes that differ in speed depending on whether stimuli are perceived as near or far from the body (Serino 2019; Bufacchi and Iannetti 2018; Karakoc et al. 2025). The utility of this task is that it uses visuotactile synchrony detection probes which remain independent of the visual intervention, and can be administered during the 4-second breath-hold windows after inhaling and exhaling, capturing shifts in self-boundaries in real time while participants remain immersed. The aim would be to test whether the system's core mapping, in which inhalation expands surrounding space and exhalation contracts it, produces corresponding expansion and contraction of the peripersonal space boundary. Reversing this mapping could test whether congruence between respiratory phase and spatial direction is what drives the effect. Demonstrating that respiratory signals can measurably reshape self-boundaries would establish breathing as a window to a new design space in VR, one in which visceral afferents serve as a medium through which experience is shaped, and how vr experiences can be further utilized to enhance the plasticity of self.

The particle system in the present experiment was used to convey the expansion and contraction of surrounding space in response to breathing, but particles can also encode visceral signals directly onto the body or its surroundings, or construct abstract visual configurations that convey affective states through movement, colour, shape, or synchrony. Our manipulation of dynamic particle symmetry through oscillator coupling explored only one such parameter and did not yield observable effects. Our lesson from this is that mapping aesthetic features of particle displays onto affective dimensions requires a more fine-grained approach than a single global variable assessed through altered-state questionnaires. Short aesthetic judgment tasks offer a tractable starting point, following swarm robotics research where trajectory smoothness, speed, synchronization, and spatial coherence map onto separable affective dimensions (Dietz et al. 2017; Santos and Egerstedt 2021). The immersive 3D particle environment affords considerably more degrees of freedom than planar swarm displays, with particles moving through three spatial dimensions under both local neighbour-coupling and global synchronization parameters, and testing specific feature combinations in controlled affective assessments would provide a more solid basis for these mappings than retrospective questionnaires after prolonged sessions. Because the medium is abstract rather than representational, the same particle field could in principle simultaneously encode multiple biosignals and serve as the basis for biofeedback displays that are intuitive to the user and conducive to the affective states the feedback is meant to support.

The present study demonstrates that the particle system was conducive to pleasant experiences during box breathing, with positive and perceptual effects comparable to prolonged facilitator-supported breathwork (Havenith et al. 2025b) at lower distress and no adverse events. Because the system is self-administered and requires only 10 minutes, it could support breathing practice in settings where access to a facilitator is limited or time is scarce. High-ventilation techniques are contraindicated for individuals vulnerable to panic, syncope, or hypocapnia (Fincham et al. 2023), and a slow-paced breath-coupled VR system offers a lower-intensity entry point into breathing as a therapeutic modality. Stroke survivors are one population where this could be relevant, given that respiratory dysfunction and sleep-disordered breathing are common after stroke and associated with poorer recovery (Patrizz et al. 2023; Seiler et al. 2019), while structured respiratory rehabilitation remains underemphasized in clinical practice (Menezes et al. 2016). Slow-paced breathing exercises are already used to retrain respiratory function after stroke (Abdullahi et al. 2024; Yoon et al. 2022), and coupling breathing to an intuitive visual mapping of respiratory depth could engage an additional learning pathway that supports recovery of motor control over respiration. These remain illustrative possibilities. The present sample consisted of a homogeneous group of university psychology students, and testing usability, tolerability, and efficacy in clinical populations is a necessary next step.

4.8 Conclusion

This study demonstrates that Viscereality, a breath-responsive VR environment that maps lung volume onto a surrounding particle field, is conducive to positive experiential states and supports guided breathing practice during brief self-administered sessions with naive participants. In 10 minutes of box breathing, the system shifted spatial self-experience toward a more expanded frame of reference and produced an altered-state profile in which positive and perceptual effects were comparable in magnitude to those reported after prolonged facilitator-supported circular breathwork, while producing less distress and no adverse events across 47 participants.

Statements and Declarations

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Competing interests: The authors declare no competing interests.

Ethics approval: The Ethics Committee of the University of Konstanz approved the study, which was conducted in accordance with the Declaration of Helsinki.

Consent to participate: All participants provided informed consent before participation.

Consent for publication: Participants consented to the publication of anonymized, non-identifiable data.

Availability of data and materials: The deidentified participant-level data, original preregistration record, statistical analysis outputs, figure-generation scripts, study materials, and reproducibility documentation are available as a reproducibility package at OSF: https://doi.org/10.17605/OSF.IO/64MWK.

Code availability: The statistical analysis and figure-generation scripts, exact environment specifications, and replay instructions are available at OSF: https://doi.org/10.17605/OSF.IO/64MWK.

Authors' contributions: George Fejer: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. Till Holzapfel: Conceptualization, Methodology, Resources, Software, Validation, Visualization. Taru Hirvonen: Methodology, Resources, Software, Visualization. Anestis Lalidis Mateo: Methodology, Resources, Software. Johannes Blum: Methodology, Resources, Software, Validation. Michael Gaebler: Conceptualization, Project administration, Supervision, Writing – review & editing. Bigna Lenggenhager: Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

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